

Geometry of Yang-Mills Moduli Space

USTC, You Jiangcheng

Contents

Chapter 1	Basic Construction for Yang-Mills Moduli Space	1
1	Moduli Space of Irreducible Self-Dual Connections	1
2	The First Example: The BPST Instanton	9
3	The Moduli Space of the Based Gauge Group	9
Chapter 2	Geometry of Yang-Mills Moduli Space for Definite Manifolds,	
	I: Geometry of the Collar	23
1	Step 1: Construct the Approximate Tangent Space	24
2	Step 2: Compute the Differential of the Collar Map	32
3	Step 3: Analyzing the Asymptotic Behavior of the Metric on the Collar . .	40
Chapter 3	Geometry of Yang-Mills Moduli Space for Definite Manifolds,	
	II: Geometry of the Cones	46
Chapter 4	Curvature of Yang-Mills Moduli Spaces	53
1	Curvature Near the Boundary	53
2	Sharp Decay Estimates for Yang-Mills Fields	54

Chapter 1 Basic Construction for Yang-Mills Moduli Space

1 Moduli Space of Irreducible Self-Dual Connections

Step1: what is Yang-mills moduli space

Let (M, g) be a compact, oriented, Riemannian 4-manifold and $P \rightarrow M$ a principal bundle whose structure group is a compact semi-simple Lie group G with Lie algebra $\mathfrak{g}$. Firstly, we can complete the space of connections. Let $\mathcal{A}$ denote the set of smooth connections on P . Then the covariant exterior derivatives of two connections $\mathcal{A}_0, \mathcal{A}$ are related by

$$d_A - d_{A_0} = P_X \quad (1.1)$$

For some $X \in \Omega^1(\text{Ad}P)$. Where the operator P_Y associated with an element $Y = \omega_i B_i \in \Omega^1(\text{Ad}P)$ is defined by

$$\begin{aligned} P_Y: \Omega^q(\text{Ad}P) &\rightarrow \Omega^{q+1}(\text{Ad}P) \\ \omega' \otimes B' &\mapsto \sum \omega_i \wedge \omega' \otimes [B_i, B'] \end{aligned}$$

If we write $X, Y \in \Omega^1(\text{Ad}P)$ in local orthonormal coframe, $X = X_i \theta^i, Y = Y_i \theta^i$, the adjoint of P_X satisfies

$$P_X^* Y = -P_Y^* X = -\sum [X_i, Y_i].$$

Hence $\mathcal{A}$ is an affine space, and at each $A \in \mathcal{A}$, there is a natural identification of the tangent space $T_A \mathcal{A}$ with $\Omega^1(\text{Ad}P)$. It is noteworthy that we treat the tangent space at each point of $\mathcal{A}$ the same, then the metric on $\mathcal{A}$ inherited from the inner product on $\Omega^*(\text{Ad}P)$ via the above identification is flat. Then the space $\mathcal{A}_\ell$ of Sobolev connections is defined by fixing $A_0 \in \mathcal{A}$, identifying $\mathcal{A}$ with $\Omega^1(\text{Ad}P)$ via (1.1), and completing in the Sobolev ℓ -norm.

Each $A \in \mathcal{A}_\ell$ determines Laplacians $\Delta_A^0 = d_A^* d_A$, $\Delta_A^1 = d_A d_A^* + (d_A^-)^* d_A^-$, $\Delta_A^2 = d_A^- (d_A^-)^*$ (where $d_A^- = p_- d_A$, and p_- is the projection onto subspace $\Omega_-^2(\text{Ad} P)$ of $\Omega^2(\text{Ad} P)$). Actually, they are elliptical. Recall A linear differential operator P of order m is elliptic if its principal symbol $\sigma(P)(\xi)$ is invertible for all non-zero cotangent vectors $\xi \neq 0$. For second-order operators, we verify positive definiteness of the symbol matrix.

- Verification for $\Delta_A^0 = d_A^* d_A$.

Domain: Acts on 0-forms (sections of $\text{Ad } P$).

Symbol computation:

$$\begin{aligned}\sigma(d_A)(\xi) &= i\xi \wedge \cdot \quad (\text{acting on 0-forms}) \\ \sigma(d_A^*)(\xi) &= -i\iota_\xi \\ \sigma(\Delta_A^0)(\xi) &= \sigma(d_A^*) \circ \sigma(d_A) = (-i\iota_\xi)(i\xi \wedge \cdot) = |\xi|^2\end{aligned}$$

Since $|\xi|^2 > 0$ for $\xi \neq 0$, Δ_A^0 is elliptic.

- Verification for $\Delta_A^1 = d_A d_A^* + (d_A^-)^* d_A^-$.

Domain: Acts on 1-forms.

Symbol decomposition:

- First term $d_A d_A^*$:

$$\sigma(d_A d_A^*)(\xi)\alpha = \xi \otimes (\xi \cdot \alpha)$$

Projects α onto ξ -direction with coefficient $|\xi|^2$.

- Second term $(d_A^-)^* d_A^-$:

$$\sigma((d_A^-)^* d_A^-)(\xi) = p_-(\xi \wedge \cdot)^* \circ p_-(\xi \wedge \cdot)$$

Acts on components orthogonal to ξ , positive definite on $\Omega_-^2(\text{Ad } P)$.

Total symbol:

$$\sigma(\Delta_A^1)(\xi) = |\xi|^2 \text{ (along } \xi) \oplus |\xi|^2 \text{ (orthogonal complement)}$$

Positive definite for $\xi \neq 0$, hence elliptic.

- Verification for $\Delta_A^2 = d_A^- (d_A^-)^*$.

Domain: Acts on $\Omega_-^2(\text{Ad } P)$.

Symbol computation:

$$\begin{aligned}\sigma(d_A^-)(\xi) &= p_-(\xi \wedge \cdot) \\ \sigma((d_A^-)^*)(\xi) &= \iota_\xi \circ p_- \\ \sigma(\Delta_A^2)(\xi)\beta &= p_-(\xi \wedge \iota_\xi \beta) = |\xi|^2 \beta \quad \forall \beta \in \Omega_-^2(\text{Ad } P)\end{aligned}$$

Positive definite on the anti-self-dual space, thus elliptic.

Thus all three operators Δ_A^0 , Δ_A^1 , and Δ_A^2 are elliptic because their principal symbols are positive definite/invertible for all $\xi \neq 0$.

By the spectral theorem for self-adjoint elliptic operators, each space $\Omega_\ell^q(\text{Ad}P)$ decomposes as the direct sum of the finite-dimensional eigenspaces of Δ_A^q (where $q = 0, 1, 2$ and $\Omega_\ell^q(\text{Ad}P)$ is the completion of $\Omega^q(\text{Ad}P)$ with respect to Sobolev norm)

$$\Omega_\ell^q(\text{Ad}P) = K^q \oplus B_\ell^q,$$

Where $K^q = \ker \Delta_A^q$ is the finite-dimensional space of Δ_A^q -harmonic forms. Furthermore, $\Delta_A^q: B_{\ell+2}^q \rightarrow B_\ell^q$ is a bounded map with a bounded inverse, or Green operator

$$G_A^q: B_\ell^q \rightarrow B_{\ell+2}^q.$$

The connection A is called **irreducible** if $K^0 = 0$, or equivalently, if $\ker(d_A: \Omega^0 \rightarrow \Omega^1) = 0$. Then Δ_A^0 is invertible on $\Omega_\ell^0(\text{Ad}P)$. Furthermore, it's not hard to show that the set of irreducible connections is an open dense subset $\widehat{\mathcal{A}}_\ell \subset \mathcal{A}_\ell$.

The set of all smooth automorphisms of principal bundle P is called the **gauge group** $\mathcal{G}$. It's well known that the gauge group $\mathcal{G}$ can be naturally identified with the space of sections of the associated bundle $\text{Aut}P = P \times_{\text{Ad}} G$. For $\ell \geq 3$, we choose a faithful representation $\rho: G \rightarrow \text{End}(V)$, then there exists an inclusion $\mathcal{G} \hookrightarrow \Gamma(P \times_\rho \text{End}(V))$, and $\mathcal{G}_\ell$ is the closure of $\mathcal{G}$ in the Sobolev ℓ -completion of $\Gamma(P \times_\rho \text{End}(V))$. Thus defined, $\mathcal{G}_\ell$ is a smooth Hilbert Lie group whose Lie algebra is $\Omega_\ell^0(\text{Ad}P)$. A gauge transformation $g \in \mathcal{G}$ acts on connections by $d_A \mapsto g \cdot d_A := g \circ d_A \circ g^{-1}$, hence sends curvature F_A to $(\text{Ad}_g)F_A$. This action extends to a smooth action of $\mathcal{G}_{\ell+1}$ on $\mathcal{A}_\ell$ whose differential at $A \in \mathcal{A}_\ell$ can be computed as follows. We write $d_A = d + A$, since

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \exp(t\xi) \circ d_A \circ \exp(-t\xi) &= \left. \frac{d}{dt} \right|_{t=0} \exp(t\xi) d \exp(-t\xi) + \exp(t\xi) A \exp(-t\xi) \\ &= -d\xi + \xi A - A\xi \\ &= -d_A \xi, \end{aligned}$$

hence the differential at $A \in \mathcal{A}_\ell$ is

$$-d_A: T_{\text{Id}} \mathcal{G}_{\ell+1} = \Omega_{\ell+1}^0(\text{Ad}P) \rightarrow T_A \mathcal{A}_\ell = \Omega_\ell^1(\text{Ad}P).$$

Consequently, the tangent space at an irreducible connection $A \in \widehat{\mathcal{A}}_\ell$ is the direct sum of subspaces

$$T_A \widehat{\mathcal{A}}_\ell = V_A \oplus H_A. \quad (1.2)$$

Where the vertical subspace $V_A = \text{Im } d_A$ is the tangent space to the gauge-orbit through A , horizontal subspace $H_A = \ker d_A^*$ is the L^2 -orthogonal complement. We will denote

the gauge-equivalence class of a connection A by $[A]$, from a standard slice theorem, we deduce that the orbit space $\mathcal{O}_\ell$ is a Hausdorff Hilbert manifold, and it's rational to identify $T_{[A]}\mathcal{O}_\ell$ with horizontal subspace H_A . More precisely, we need the Slice Theorem Framework: (Observe that the irreducibility ensures free gauge group action and smooth Hausdorff structure on $\mathcal{O}_\ell$ without singular point.)

Theorem 1.1 (Slice Theorem for Gauge Actions). *Let $\mathcal{G}_{\ell+1}$ act smoothly and freely on the Sobolev connection space $\mathcal{A}_\ell$. For any irreducible connection $A \in \widehat{\mathcal{A}}_\ell$, there exists a local slice $S_A \subset \mathcal{A}_\ell$ such that:*

1. S_A is transverse to the gauge orbit $\mathcal{G}_{\ell+1} \cdot A$;
2. The tangent space satisfies $T_A S_A = H_A$;
3. A neighborhood of $[A]$ in $\mathcal{O}_\ell$ is diffeomorphic to S_A .

The slice theorem provides a local submanifold $S_A \subset \mathcal{A}_\ell$ with $T_A S_A = H_A$. Moreover, since there exists a diffeomorphism between S_A and a neighborhood of $[A]$ in $\mathcal{O}_\ell$, we have

$$T_{[A]}\mathcal{O}_\ell \cong T_A S_A = H_A.$$

Moreover, the L^2 -orthogonal splitting (1.2) determines vertical and horizontal projections v_A, h_A at each $A \in \widehat{\mathcal{A}}_\ell$. Obviously, for any $X \in \Omega^1(\text{Ad}P)$, we write

$$X = d_A G_A^0 d_A^* X + (X - d_A G_A^0 d_A^* X).$$

Since the first term is in $\text{Im} d_A$ and the second term is in $\ker d_A^*$, we have

$$v_A = d_A G_A^0 d_A^*, \quad h_A = \text{Id} - d_A G_A^0 d_A^*. \quad (1.3)$$

Now we can introduce the self-dual moduli space, the image of the set of all self-dual $\text{SU}(2)$ connections in the orbit space as follow

$$\mathcal{M} = \{[A] : A \text{ is self dual}\} \in \mathcal{O}_\ell.$$

It contains an open dense subset, the moduli space

$$\widehat{\mathcal{M}} = \{[A] : A \text{ is self dual and } \ker \Delta_A^0 = 0\} \in \mathcal{O}_\ell$$

of irreducible self-dual connections.

Step2: Is that possible moduli space be a manifold?

Atiyah proved in his paper, “*Self-duality in four dimensional Riemannian Geometry*”, that when nonempty, the following subset

$$\widehat{\mathcal{M}} = \{[A] \in \widehat{\mathcal{M}} : \ker \Delta_A^2 = 0\}$$

is a manifold of dimension $2p_1(\text{Ad}P)[M] - \dim G(1 - b_1 + b_2^-)$, where $b_1 = \dim(H^1(M; \mathbb{R}))$ and b_2^- is the dimension of the space of ASD harmonic 2-forms on M . Now the remained difficulty is the connections A satisfied $\ker \Delta_A^2 \neq 0$. However, a vanishing theorem based on the Weitzenböck formula claims that, under a certain curvature condition on M , the kernel of Δ_A^2 is 0 for any connection A in the moduli space. Henceforth, we will treat $\widehat{\mathcal{M}}$ as a manifold.

Step3: As a manifold, can we compute curture of $\widehat{\mathcal{M}}$?

From now on, we fix a Sobolev index ℓ and denote $\mathcal{A}_\ell, \mathcal{G}_\ell, \mathcal{O}_\ell$ by $\mathcal{A}, \mathcal{G}, \mathcal{O}$ respectively. Before computing the curvature of $\mathcal{M}_1$, we need to describe the metric on it. Firstly, the affine space $\mathcal{A}$ inherits a weak Riemannian metric from the L^2 inner product on $\Omega^*(\text{Ad}P)$ via the identification $T_A \mathcal{A} = \Omega^1(\text{Ad}P)$. This metric is flat. The group $\mathcal{G}$ acts isometrically on $\mathcal{A}$ and, at each $A \in \widehat{\mathcal{A}}$, preserves the splitting (1.2). Therefore, we can give the orbit space $\mathcal{O}$ a weak Riemannian metric by identifying $T_{[A]} \mathcal{O}_\ell$ with horizontal subspace H_A . To be specific, if A_1 and A_2 are in the same equivalence class, the precisely one tangent space has two forms of metric, induced from H_{A_1} and H_{A_2} . We check the L^2 -metric induced by above differential ways coincides.

The horizontal subspaces transform under g as:

$$H_{A_2} = \ker d_{A_2}^* = g \cdot \ker d_{A_1}^* = g \cdot H_{A_1}.$$

The gauge group $\mathcal{G}$ acts isometrically on $\mathcal{A}$, preserving the L^2 -inner product implies

$$\langle g \cdot \alpha, g \cdot \beta \rangle_{L^2} = \langle \alpha, \beta \rangle_{L^2} \quad \forall \alpha, \beta \in T_A \mathcal{A}.$$

Thus for any horizontal vectors $\alpha_1, \beta_1 \in H_{A_1}$, define $\alpha_2 = g \cdot \alpha_1$, $\beta_2 = g \cdot \beta_1 \in H_{A_2}$, we have

$$\langle \alpha_2, \beta_2 \rangle_{L^2(A_2)} = \langle \alpha_1, \beta_1 \rangle_{L^2(A_1)}.$$

Hence the inner product is preserved under gauge transformation. Recall the metric on $T_{[A]} \mathcal{O}$ is defined via horizontal representatives:

$$\langle [\alpha], [\beta] \rangle_{T_{[A]} \mathcal{O}} = \langle \alpha_1, \beta_1 \rangle_{L^2(A_1)} = \langle \alpha_2, \beta_2 \rangle_{L^2(A_2)}.$$

This is independent of the choice of A_1 or A_2 in the orbit.

Finally, the moduli space $\widehat{\mathcal{M}}$ is an embedding submanifold of $\mathcal{O}$, then there is an induced Riemannian metric on $\widehat{\mathcal{M}}$.

In this part, we will compute the curvature of this L^2 induced metric on $\widehat{\mathcal{M}}$ via the following diagram

$$\begin{array}{ccc} & \widehat{\mathcal{A}} & \\ & \downarrow \pi & \\ \widehat{\mathcal{M}} & \longrightarrow & \mathcal{O} = \widehat{\mathcal{A}}/\mathcal{G} \end{array}$$

Cheeger and Ebin's book, "Comparison theorems in Riemannian geometry" showed that π **actually is a Riemannian submersion**, this allow us to apply O'Neil's formula to compute the curvature of $\mathcal{O}$.

Now we describe the Levi-Civita connection on $\widehat{\mathcal{A}}$ and $\mathcal{O}$. The argument of existence and uniqueness of connection on finite-dimensional manifolds is still vaild in infinite dimensions provided each tangent space is a Hilbert space with respect to the Riemannian metric. However, in our case, tangent spaces of $\widehat{\mathcal{A}}$ and $\mathcal{O}$ are not complete with respect to the L^2 metric. For this L^2 metric, the uniqueness of Levi-Civita connection can be guaranteed, but not its existence. We will exhibiting an explict formula for connection ∇ to handle existence as follow.

Using the identification $T_A\mathcal{A} = \Omega^1(\text{Ad}P)$, we regard a vector field Y on an open subset of $\mathcal{A}$ as a map $Y: U \rightarrow \Omega^1(\text{Ad}P)$ and set

$$(\nabla_X^{\mathcal{A}} Y)_A = \left. \frac{d}{dt} \right|_{t=0} Y(A + tX(A)) \in \Omega^1(\text{Ad}P). \quad (1.4)$$

This defines a connection that is torsion-free and compatible with the L^2 metric. We check as below:

- Torsion-free. Compute directly,

$$\begin{aligned} (\nabla_X^{\mathcal{A}} Y - \nabla_Y^{\mathcal{A}} X)_A &= \left. \frac{d}{dt} \right|_{t=0} Y(A + tX(A)) - \left. \frac{d}{dt} \right|_{t=0} X(A + tY(A)) \\ &= DY_A(X(A)) - DX_A(Y(A)) \\ &= [X, Y]_A. \end{aligned}$$

- Compatible with L^2 metric. The L^2 -inner product is:

$$\langle Y, Z \rangle_A = \int_M \text{Tr}(Y(A) \wedge \star Z(A)).$$

Left-hand side

$$X \langle Y, Z \rangle = \left. \frac{d}{dt} \right|_{t=0} \int_M \text{Tr}(Y(A + tX(A)) \wedge \star Z(A + tX(A))).$$

Right-hand side

$$\begin{aligned}
 \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle &= \int_M \text{Tr} \left(\frac{d}{dt} \Big|_{t=0} Y(A + tX(A)) \wedge \star Z(A) \right) \\
 &\quad + \int_M \text{Tr} \left(Y(A) \wedge \star \frac{d}{dt} \Big|_{t=0} Z(A + tX(A)) \right) \\
 &= \frac{d}{dt} \Big|_{t=0} \int_M \text{Tr} (Y(A + tX(A)) \wedge \star Z(A + tX(A)))
 \end{aligned}$$

By linearity of differentiation and the product rule, both sides are equal.

Using formula (1.4), we derive a Levi-Civita connection on $\mathcal{O}$. Specifically, given $X \in T_{[A]}\mathcal{O}$ and a vector field defined in a neighbourhood of $[A]$, we choose horizontal lifts $\bar{X}, \bar{Y}$ of X, Y , which means $\pi_* \bar{X} = X, \pi_* \bar{Y} = Y$ hold, and set

$$(\nabla_X^\mathcal{O} Y)_A = \pi_*(\nabla_{\bar{X}}^\mathcal{A} \bar{Y})_A. \quad (1.5)$$

For Riemannian Submersion π , the curvatures of $\widehat{\mathcal{A}}$ and $\mathcal{O}$ are related by O'Neil's formula

$$\langle R_{\text{down}}(X_p, Y_p)Y_p, X_p \rangle = \langle R_{\text{up}}(\bar{X}_q, \bar{Y}_q)\bar{Y}_q, \bar{X}_q \rangle + \frac{3}{4} \|\text{vert}[\bar{X}, \bar{Y}]_q\|^2, \quad (1.6)$$

where $[\bar{X}, \bar{Y}]_q$ is the Lie bracket of arbitrary horizontal extensions $\bar{X}, \bar{Y}$ of X_q, Y_q , and “vert” denotes the vertical projection onto the vertical subspace. The inherited weak Riemannian metric on $\mathcal{A}$ is flat, thus we only need to compute the second term in RHS of (1.6).

Now fix $A_0 \in \widehat{\mathcal{A}}$ and $X_0, Y_0 \in H_{A_0}$, we choose horizontal projections of X_0, Y_0 as horizontal extensions of them, if X_0, Y_0 is viewed as constant vector fields. i.e.

$$\bar{X}_A = h_A X_0; \quad \bar{Y}_A = h_A Y_0 \quad (1.7)$$

By (1.4) and formula (1.3) for h_A , we have

$$\begin{aligned}
 (\nabla_{X_0}^\mathcal{A} \bar{Y})_{A_0} &= \frac{d}{dt} \Big|_{t=0} \bar{Y}_{A_0+tX_0} \\
 &= \frac{d}{dt} \Big|_{t=0} h_{A_0+tX_0} Y_0 \\
 &= -\frac{d}{dt} \Big|_{t=0} \left(d_{A_0+tX_0} G_{A_0+tX_0}^0 d_{A_0+tX_0}^* Y_0 \right)
 \end{aligned}$$

Thanks to $d_{A_0}^* Y_0 = 0$ and $d_{A_0+tX_0}^* - d_{A_0}^* = P_X^*$, the above consequence can be simplified,

$$(\nabla_{X_0}^\mathcal{A} \bar{Y})_{A_0} = -d_{A_0} G_{A_0}^0 P_{X_0}^* Y_0, \quad (1.8)$$

$\nabla^\mathcal{A}$ is torsion free, hence

$$[\bar{X}, \bar{Y}]_{A_0} = \left(\nabla_{X_0}^\mathcal{A} \bar{Y} - \nabla_{Y_0}^\mathcal{A} \bar{X} \right)_{A_0} = -2d_{A_0} G_{A_0}^0 P_{X_0}^* Y_0. \quad (1.9)$$

Observe that this is already vertical, therefore

$$\begin{aligned} \|\text{vert}[\bar{X}, \bar{Y}]_{A_0}\|^2 &= 4\|d_{A_0}G_{A_0}^0P_{X_0}^*Y_0\|^2 \\ &= 4\langle d_{A_0}^*d_{A_0}G_{A_0}^0P_{X_0}^*Y_0, G_{A_0}^0P_{X_0}^*Y_0 \rangle \\ &= 4\langle P_{X_0}^*Y_0, G_{A_0}^0P_{X_0}^*Y_0 \rangle \end{aligned}$$

Putting this into formula (1.6) yields

$$\langle R_{\mathcal{O}}(X, Y)Y, X \rangle = 3\langle P_{\bar{X}}^*\bar{Y}, G_{A_0}^0P_{\bar{X}}^*\bar{Y} \rangle, \quad (1.10)$$

where $X, Y \in T_{[A]}\mathcal{O}$ and $\bar{X}, \bar{Y}$ project to X, Y . Eventually, since Riemannian curvature can be expressed by sectional curvature, we obtain the following consequence

Theorem 1.2. *The curvature of the L^2 metric on $\mathcal{O}$ at a point $[A]$ is given by*

$$\langle R_{\mathcal{O}}(X, Y)Z, W \rangle = \langle P_{\bar{X}}^*\bar{W}, G_A^0P_{\bar{Y}}^*\bar{Z} \rangle - \langle P_{\bar{Y}}^*\bar{W}, G_A^0P_{\bar{X}}^*\bar{Z} \rangle + 2\langle P_{\bar{W}}^*\bar{Z}, G_A^0P_{\bar{X}}^*\bar{Y} \rangle \quad (1.11)$$

where $\bar{X}, \bar{Y}, \bar{Z}, \bar{W}$ are the horizontal lifts to $A \in \widehat{\mathcal{A}}$ of $X, Y, Z, W \in T_{[A]}\mathcal{O}$.

Now we compute the curvature of moduli space $\widehat{\mathcal{M}}$ based on the fact, $\widehat{\mathcal{M}}$ isometrically imbedding in $\mathcal{O}$. The second fundamental form of the embedding $\iota: \widehat{\mathcal{M}} \rightarrow \mathcal{O}$ is a section b of $\text{Sym}^2(T^*\widehat{\mathcal{M}}) \otimes \nu$, where ν is the normal bundle. The form b is defined as follow

$$b(X, Y) = (\nabla_{d_X}^\mathcal{O} d_Y)^\perp \quad (1.12)$$

Where $X, Y \in T\widehat{\mathcal{M}}$ and $\perp$ denotes the L^2 -orthogonal projection from $T\mathcal{O}$ onto ν . Moreover, this projection is well-defined since each $T_{[A]}\widehat{\mathcal{M}}$ is finite-dimensional. Now we compute $b(X, Y)$. Fix $[A_0] \in \widehat{\mathcal{M}}$ and consider the slice $S_0 = A_0 + \ker d_{A_0}^* \subset \mathcal{A}$. The manifold $N = \pi^{-1}\widehat{\mathcal{M}} \cap S_0$ passes through A_0 and $\pi: N \rightarrow \widehat{\mathcal{M}}$ is a local diffeomorphism. Furthermore, $T_{A_0}N$ is the subspace $T_0 = \ker d_{A_0}^* \cap \ker d_{A_0}^-$ of $\Omega_\ell^1(\text{Ad } P)$, and the normal bundle ν pulls back to a bundle over N whose fiber at A_0 is

$$\nu_0 = \{Z \in \ker d_{A_0}^* : Z \perp T_0 \subset T_{A_0}\mathcal{A}\}$$

Now given $X_0, Y_0 \in T_0$, we can construct their horizontal extensions as follows

$$X_A^{\text{tan}} = (1 - (d_A^-)^*G_A^2d_A^-)X_A$$

Then we obtain horizontal vector fields $X^{\text{tan}}, Y^{\text{tan}}$ whose projections $d\pi X^{\text{tan}}, d\pi Y^{\text{tan}}$ are tangengt to $\widehat{\mathcal{M}}$. Computing as we did to arrive at (1.8), noting that $d_{A_0}^-Y_0 = 0$ and $d_{A+tX}^- = d_A^- + P_X^-$, we have

$$(\Delta_{X_0}^\mathcal{A} Y^{\text{tan}})_{A_0} = -(d_{A_0}^-)^*G_{A_0}^2P_{X_0}^-Y_0 - d_{A_0}G_{A_0}^0P_{X_0}^*Y_0 \in T_{A_0}\mathcal{A}.$$

The last term is vertical, so it drops out upon projection to $\mathcal{O}$. Meanwhile, the remaining piece is already in the normal space $\text{Im } d_{A_0}^*$, hence from (1.5), (1.12) we deduce

$$b(X, Y)_A = -(d_A^-)^* G_A^2 P_X^- \bar{Y}. \quad (1.13)$$

Where $\bar{X}, \bar{Y}$ are horizontal lifts of vectors X, Y on $\mathcal{M}$. Now from theorem 1.2, Gauss equation and (1.13), we obtain the following theorem,

Theorem 1.3. *The curvature of the moduli space $\widehat{\mathcal{M}}$ at a point $[A]$ is given by*

$$\begin{aligned} \langle R_{\widehat{\mathcal{M}}}(X, Y)Z, W \rangle &= \langle P_X^* \bar{W}, G_A^0 P_Y^* \bar{Z} \rangle - \langle P_Y^* \bar{W}, G_{A_0}^0 P_X^* \bar{Z} \rangle + 2 \langle P_W^* \bar{Z}, G_{A_0}^0 P_X^* \bar{Y} \rangle \\ &\quad + \langle P_Y^- \bar{Z}, G_A^2 P_X^- \bar{W} \rangle - \langle P_X^- \bar{Z}, G_A^2 P_Y^- \bar{W} \rangle \end{aligned} \quad (1.14)$$

where $\bar{X}, \bar{Y}, \bar{Z}, \bar{W}$ are the horizontal lifts to $A \in \widehat{\mathcal{A}}$ of $X, Y, Z, W \in T_{[A]} \widehat{\mathcal{M}}$. In particular, the sectional curvatures are given by

$$\langle R_{\widehat{\mathcal{M}}}(X, Y)Y, X \rangle = 3 \langle P_X^* \bar{Y}, G_A^0 P_X^* \bar{Y} \rangle + \langle P_Y^- \bar{Z}, G_A^2 P_X^- \bar{W} \rangle - \langle P_X^- \bar{Z}, G_A^2 P_Y^- \bar{W} \rangle \quad (1.15)$$

2 The First Example: The BPST Instanton

3 The Moduli Space of the Based Gauge Group

Different from irreducible connections, reducible connections prevent the action of the gauge group $\mathcal{G}$ on $\mathcal{A}$ from being locally free, and consequently the orbit space $\mathcal{B}$ and the moduli space $\mathcal{M} \subset \mathcal{B}$ are usually not manifolds. To some extent, we may treat the orbits through reducible connections as singular points of $\mathcal{M}$. This section requires no assumptions on M , P , and G , save that the 4-manifold M be compact and oriented, and that the Lie group G be compact and semisimple.

Fix a basepoint $x_0 \in M$ and considering the normal subgroup $\mathcal{G}^0 \subset \mathcal{G}$ consisting of those automorphisms of P which restrict to the identity on P_{x_0} , the fiber over x_0 . This “based” gauge group acts freely on $\mathcal{A}$, so $\mathcal{A} \rightarrow \tilde{\mathcal{B}} = \mathcal{A}/\mathcal{G}^0$ is a principal $\mathcal{G}^0$ -bundle. On the other hand, the full gauge group $\mathcal{G}$ acts on $\mathcal{A}$ and the stabilizer at each irreducible connection is the center of $\mathcal{G}$, i.e. $\text{Stab}_{\mathcal{G}}(A) = \mathcal{Z}$ (which be commonly referred to as the definition of irreducible connection). This center is the group $\mathcal{Z} = \Gamma(P \times_{\text{Ad}} Z)$, where Z is the center of G ($\mathcal{Z} \cong Z$ is finite since G is semisimple). Thus the action of $\mathcal{G}$ on $\mathcal{A}$ induces an action of $\mathcal{G}/(\mathcal{Z} \times \mathcal{G}^0) \cong G/Z$ on $\tilde{\mathcal{B}}$ which is free on the open dense subset $\tilde{\mathcal{B}}^*$ of irreducible connections (which will be proved in lemma 1.4), and whose orbit space is $\mathcal{B} = \mathcal{A}/\mathcal{G}$. Restricting to the self-dual connections $\mathcal{SD} \subset \mathcal{A}$, we get a moduli space $\widehat{\mathcal{M}} \subset \tilde{\mathcal{B}}$ with a G/Z -action whose orbit space is the usual moduli space $\mathcal{M} = \mathcal{SD}/\mathcal{G} \subset \mathcal{B}$, that is

$$(\mathcal{SD}/\mathcal{G}^0)/(G/Z) \cong \mathcal{SD}/\mathcal{G}.$$

In summary, the quotient space of self-dual connections is constructed in two stages:

$$\mathcal{SD} \xrightarrow{\text{quotient by } \mathcal{G}^0} \widetilde{\mathcal{M}} = \mathcal{SD}/\mathcal{G}^0 \xrightarrow{\text{quotient by } G/Z} \mathcal{M} = \mathcal{SD}/\mathcal{G}. \quad (1.16)$$

And the relationships across all spaces can be described collectively as

$$\begin{array}{ccc} \widetilde{\mathcal{M}} & \hookrightarrow & \widetilde{\mathcal{B}} \\ \text{quotient by } G/Z \downarrow & & \downarrow \text{quotient by } G/Z \\ \mathcal{M} & \hookrightarrow & \mathcal{B} \end{array} \quad (1.17)$$

Let us now interpret the meaning of the aforementioned construction. Based gauge group $\mathcal{G}^0$ plays a crucial role. Firstly, $\mathcal{G}^0$ acts freely on the configuration space $\mathcal{A}$ (without fixed points), so the quotient space $\widetilde{\mathcal{B}} = \mathcal{A}/\mathcal{G}^0$ is a smooth principal $\mathcal{G}^0$ -bundle. This makes $\widetilde{\mathcal{B}}$ have a good fibrous structure.

(1.16) actually illustrated the separation of gauge group redundancy. More explicitly, we decomposed the gauge group action $\mathcal{G}$ on $\mathcal{SD}$ into two steps:

- Step 1: Eliminate the “base-point-based” local redundancy through $\mathcal{G}^0$,
- Step 2: The remaining finite group G/Z acts on $\widetilde{\mathcal{M}}$, and its orbit space is the final moduli space $\mathcal{M} = \mathcal{SD}/G$. Fixed points of Group action G/Z on $\widetilde{\mathcal{M}}$ correspond to reducible connections, leading to singularities, such as conical singularities or orbifold singularities in $\mathcal{M}$.

This two steps decomposes the gauge redundancy into local $\mathcal{G}^0$ and global G/Z parts, which are handled separately. It is worth noting that through the fibration $\widetilde{\mathcal{M}} \rightarrow \mathcal{M}$, we separated the irreducible part (smooth) from the reducible part (singularities). Consequently, we may identify singularities arising from the fixed points of G/Z (reducible connections), rather than from $\mathcal{G}^0$, that simplified the following process of geometric analysis.

In summary, **The singularities of $\mathcal{M}$ and $\mathcal{B}$ can then be described by studying this G/Z -action.**

Lemma 1.4. *Above elaboration involved the following facts*

- 1) $\mathcal{G}/(\mathcal{L} \times \mathcal{G}^0) \cong G/Z$.
- 2) $\text{Stab}_{G/Z}([A]) = \{e\}$ for irreducible connection A .

Proof. For part 1), the proof proceeds by constructing a surjective homomorphism and analyzing its kernel. Define the homomorphism:

$$\phi : \mathcal{G} \rightarrow G, \quad g \mapsto g(x_0),$$

where $g(x_0) \in G$ is the value of the gauge transformation g at the basepoint x_0 . We claim that ϕ is surjective and the kernel of ϕ consists of gauge transformations that act as the identity map at x_0 .

To check ϕ is surjective, near the base point x_0 , we choose a local trivialization neighborhood $U \subset M$ such that $P|_U \cong U \times G$, and take a smooth cutoff function $\rho : M \rightarrow [0, 1]$ satisfying $\rho(x_0) = 1$ and $\rho(x) = 0$ outside U . In local coordinates, define the gauge transformation $g \in \mathcal{G}$ as:

$$g(x) = \begin{cases} \rho(x) \cdot g_0 + (1 - \rho(x)) \cdot e, & x \in U \\ e, & x \notin U \end{cases}$$

where e is the identity element of G . This definition gives $g(x_0) = g_0$ and maintains smoothness globally.

To check $\ker \phi = \mathcal{G}^0$, on the one hand,

$$\begin{aligned} g \in \mathcal{G}^0 &\Rightarrow \phi(g) = g(x_0) = g(x_0, h) = (x_0, h) = (x_0, g(x_0)h) \\ &\Rightarrow g(x_0) = e. \end{aligned}$$

Where we use two different definitions of gauge transformation g . On the other hand,

$$g \in \ker \phi \Rightarrow \forall p \in P_{x_0}, g(p) = g(x_0, h) = (x_0, g(x_0)h) = (x_0, h) = g(p).$$

Above reads $\ker \phi = \mathcal{G}^0$. By the First Isomorphism Theorem:

$$\mathcal{G}/\mathcal{G}^0 \cong G.$$

Now consider the composite map:

$$\mathcal{G} \xrightarrow{\phi} G \xrightarrow{\pi} G/Z,$$

where $\pi : G \rightarrow G/Z$ is the natural projection. The kernel of the composite map $\pi \circ \phi$ is:

$$\ker(\pi \circ \phi) = \{g \in \mathcal{G} \mid \phi(g) \in Z\} = \{g \in \mathcal{G} \mid g(x_0) \in Z\}.$$

The center subgroup $\mathcal{Z} \subset \mathcal{G}$ consists of constant central gauge transformations:

$$\begin{aligned} \mathcal{Z} &= \{g \in \mathcal{G} \mid g \cdot g' = g' \cdot g \text{ for all } g' \in \mathcal{G}\} \\ &= \{g \in \mathcal{G} \mid g(x) \cdot g'(x) = g'(x) \cdot g(x) \text{ for all } x \in M, g' \in \mathcal{G}\} \\ &= \{g \in \mathcal{G} \mid g(x) = z \in Z \text{ for all } x \in M\} \end{aligned}$$

Note that G is semisimple Lie group, thus its central subgroup Z is discrete, which means $g : M \rightarrow Z$ is a constant map. It's obviously that $\mathcal{Z} \cap \mathcal{G}^0 = \{\text{Id}\}$, base on this observation, we claim: any $g \in \ker(\pi \circ \phi)$ satisfies $g(x_0) \in Z$, and such g can be decomposed uniquely:

$$g = z \cdot h, \quad \text{where } z \in \mathcal{Z}, h \in \mathcal{G}^0.$$

Indeed, we put $z \in \mathcal{Z} : z(x) = g(x_0)$ and $h \in \mathcal{G}^0 : h(x) = z^{-1}(x) \cdot g(x)$, then $g = z \cdot h$ holds. Suppose there exists another decomposition $g = z' \cdot h'$, then

$$z \cdot h = z' \cdot h' \Rightarrow (z')^{-1} \cdot z = h' \cdot h^{-1},$$

Since $(z')^{-1} \cdot z \in \mathcal{Z}$ and $h' \cdot h^{-1} \in \mathcal{G}^0$, the observation $\mathcal{Z} \cap \mathcal{G}^0 = \{\text{Id}\}$ implies that $z = z'$ and $h = h'$. Hence any $g \in \ker(\pi \circ \phi)$ can be decomposed uniquely as $g = z \cdot h$, where $z \in \mathcal{Z}$, $h \in \mathcal{G}^0$.

Thus, the kernel of $\pi \circ \phi$ is $\mathcal{Z} \times \mathcal{G}^0$, and the First Isomorphism Theorem implies

$$\mathcal{G}/(\mathcal{Z} \times \mathcal{G}^0) \cong \text{Im}(\pi \circ \phi) = G/Z.$$

As for part 2), this fact implies that the action of $\mathcal{G}/(\mathcal{Z} \times \mathcal{G}^0) \cong G/Z$ on $\tilde{\mathcal{B}}^*$ is free.

Let $[A] \in \mathcal{B}^*$ be an equivalence class of irreducible connections.

For an irreducible connection A , the stabilizer under the full gauge group $\mathcal{G}$ is the center subgroup:

$$\text{Stab}_{\mathcal{G}}(A) = \mathcal{Z},$$

where $\mathcal{Z} = \{g \in \mathcal{G} \mid g(x) = z \ \forall x \in M\}$.

Recall the quotient group G/Z acts on $\mathcal{B}^*$ via

$$[g] \cdot [A] = [\tilde{g} \cdot A], \quad [g] \in G/Z, \quad \tilde{g}(x) \equiv g, \quad [A] \in \mathcal{B}^*.$$

Suppose $[g] \in G/Z$ stabilizes $[A]$, i.e. $[g] \cdot [A] = [A]$, then there exists $h \in \mathcal{G}^0$ such that $\tilde{g} \cdot A = h \cdot A$. This implies:

$$h^{-1}\tilde{g} \cdot A = A \implies h^{-1}\tilde{g} \in \text{Stab}_{\mathcal{G}}(A) = \mathcal{Z}.$$

Hence, $h^{-1}\tilde{g} = z \in \mathcal{Z}$. Observe that $\tilde{g} \equiv g \in G$ and $z \equiv z_0 \in Z$ are constant, hence $h \equiv e$ is constant either ($h \in \mathcal{G}^0$ implies $h \in \ker(\phi)$). In the quotient G/Z , this gives:

$$[g] = [z_0] = e_{G/Z}.$$

Therefore, $\text{Stab}_{G/Z}([A]) = \{e\}$, proving the action is free. □

A detailed analysis of the construction of the aforementioned moduli space will now be presented from an analytical perspective. There are standard “slice theorems” which show that $\mathcal{M}^*$ and $\mathcal{B}^*$ are Hilbert manifolds, our next objective is that revealing $\tilde{\mathcal{M}}$ and $\tilde{\mathcal{B}}$ are manifolds.

A closer examination reveals that the usual method for obtaining a slice of the $\mathcal{G}$ -action, which makes use of the L^2 inner product on $T\mathcal{A}$, will not work for $\mathcal{G}^0$, essentially because $\mathcal{G}^0$ is not a closed subgroup in the L^2 topology. **However, $\mathcal{G}^0$ is closed after completing in an appropriate Sobolev topology, and if we use this Sobolev**

metric to define slices, we do get a slice theorem. This yields smooth structures on $\widetilde{\mathcal{M}}$ and $\widetilde{\mathcal{B}}$, and the vertical arrows in (1.17) become Riemannian submersions with respect to the Sobolev Riemannian metrics on $\widetilde{\mathcal{M}}$ and $\widetilde{\mathcal{B}}$. Furthermore, with our specific choice of Sobolev norm, the Sobolev metric and the L^2 metric on $\widetilde{\mathcal{M}}$ are equal. This key fact will enable us to use the Sobolev metric on $\widetilde{\mathcal{M}}$ to obtain information about the L^2 geometry of $\mathcal{M}$ near the singular points.

The remainder of this section is dedicated to give the Sobolev metrics on the appropriate spaces, and carry out the details of Hilbert space structure. Naturally, the essential technique is constructing the slice (a local subspace satisfying certain properties) based on the following theorem

Theorem 1.5 (Silce Theorem). *Suppose G is a group acting on a manifold M , $\Gamma = \{(x, g \cdot x) : x \in M, g \in G\} \subseteq M \times M$ is the graph of the action, and M/G is the orbit space with the quotient topology. Then*

- 1) M/G is Hausdorff if and only if Γ is closed.
- 2) If through each $x \in M$ we can find a slice of the action—that is, an open submanifold N containing x such that

$$(a) \quad T_y M = T_y(G \cdot y) \oplus T_y N \text{ for all } y \in N,$$

$$(b) \quad \text{the restriction of the projection } M \rightarrow M/G \text{ to } N \text{ is one-to-one,}$$

and if the action of G is free, then M/G is a manifold (which is Hausdorff if 1) is satisfied).

We begin by recalling the fundamental elliptic sequence (which is a complex if and only if A is self-dual)

$$0 \rightarrow \Omega^0(\text{Ad}P) \xrightarrow{d_A} \Omega^1(\text{Ad}P) \xrightarrow{d_A^-} \Omega_-^2(\text{Ad}P) \rightarrow 0,$$

The Laplacians $\square_A^0 = d_A^* d_A + \text{Id}$ and $\square_A^1 = d_A d_A^* + 2(d_A^-)^* d_A^- + \text{Id}$ are invertible for each $A_0 \in \mathcal{A}$. Fix a smooth connection A_0 and for each integer $s \geq 0$ we define the Sobolev s -norm on $\Omega^1(\text{Ad}P)$ by

$$\langle \phi, \eta \rangle_{s; A_0} = \langle \phi, (\square_{A_0}^1)^s \eta \rangle_{L^2}. \quad (1.18)$$

This definition extends to all real s as follows. By the generalized Hodge theorem there is a complete orthonormal basis $\{\phi_k\}$ of $L^2(T^*M \otimes \text{Ad}P)$ where each ϕ_k satisfies $\square_{A_0}^1 \phi_k = \lambda_k \phi_k$ for some real positive eigenvalue λ_k . Hence we can expand any $\phi \in \Omega^1(\text{Ad}P)$ as $\sum a_k \phi_k$ and define

$$\|\phi\|_{s; A_0}^2 = \sum \lambda_k^s a_k^2. \quad (1.19)$$

Let $\Omega_s^1(\text{Ad}P)$ denote the completion of $\Omega^1(\text{Ad}P)$ in the Sobolev s -norm. By identifying $\mathcal{A} = T_{A_0}\mathcal{A} = \Omega^1(\text{Ad}P)$, we obtain a complete space of connections $\mathcal{A}_s$. One can similarly complete the gauge group to a group $\mathcal{G}_{s+1}$. The results of Uhlenbeck's paper, "*Connections with L^p bounds on curvature*" imply that, for $s > 1$, $\mathcal{G}_{s+1}$ is a smooth Lie group acting smoothly on $\mathcal{A}_s$, and that the topologies on these spaces are independent of the choice of the connection A_0 used to define them. Furthermore, each $g \in \mathcal{G}_{s+1}$ is continuous (as a section of $P \times_{\text{Ad}} G$) and $\mathcal{G}_{s+1}^0 = \{g \in \mathcal{G}_{s+1} | g(x_0) = \text{Id}\}$ is a closed Lie subgroup of $\mathcal{G}_{s+1}$. The Lie algebra of $\mathcal{G}_{s+1}^0$ is $\mathfrak{g}_{s+1}^0 = \{X \in \Omega_{s+1}^0(\text{Ad}P) | X(x_0) = 0\}$. Analogously, we define the L_{s+1}^2 metric on $\Omega^0(\text{Ad}P)$ to (1.18), with $(\square_A^0)^{s+1}$ replacing $(\square_A^1)^s$.

We can eliminate the special role of the connection A_0 by introducing the natural L_s^2 Riemannian metric on $\mathcal{A}_s$. It is defined at $A \in \mathcal{A}$ by replacing A_0 in (1.18) or (1.19) by A . The Sobolev inequalities imply that for $s \geq 1$ the norms $\|\cdot\|_{s;A}$ on $\Omega^1(\text{Ad}P)$ are all locally uniformly equivalent and that $\|\cdot\|_s$ is a smooth Riemannian metric on $\mathcal{A}_s$.

The space $\mathcal{A}_s$ thus carries both a strong L_s^2 metric and a weak L^2 metric. Each defines a slice for the action of $\mathcal{G}_{s+1}$. Since the infinitesimal action of $\mathcal{G}_{s+1}$ at $A \in \mathcal{A}$ is the map $d_A: \Omega_{s+1}^0(\text{Ad}P) \rightarrow \Omega_s^1(\text{Ad}P)$, the L^2 slice H_A is defined as the L^2 -orthogonal complement of $\text{Im}(d_A)$:

$$H_A = \{\eta \in \Omega_s^1(\text{Ad}P) \mid d_A^* \eta = 0\}. \quad (1.20)$$

Similarly, the L_s^2 slice H_A^s , as the L^2 -orthogonal complement of $\text{Im}(d_A)$, is defined by

$$H_A^s = \{\eta \in \Omega_s^1(\text{Ad}P) \mid \langle d_A X, \eta \rangle_s = \langle X, d_A^*(\square_A^1)^s \eta \rangle_{L^2} = 0, \forall X\}. \quad (1.21)$$

When $F_A^- = 0$ we have $d_A^* \square_A^1 = \square_A^0 d_A^*$, so that (1.21) is equivalent to $\langle X, (\square_A^0)^s d_A^* \eta \rangle_{L^2} = 0$; i.e. $H_A^s = \ker((\square_A^0)^s \circ d_A^*)$. Because $\square_A^0$ is invertible, this simplifies to

$$H_A^s = \ker(d_A^*).$$

Thus the slices H_A and H_A^s coincide (independently of s) when A is self-dual.

Now we describe the corresponding slices for the action of the based gauge group $\mathcal{G}^0$. We consider the L_s^2 slice $\tilde{H}_A^s$ for the action of $\mathcal{G}_{s+1}^0$ on $\mathcal{A}_s$. As above, $\eta \in \tilde{H}_A^s$ if and only if η satisfies $\langle d_A X, \eta \rangle_s = \langle X, d_A^*(\square_A^1)^s \eta \rangle_{L^2} = 0$ for all $X \in \mathfrak{g}_{s+1}^0$, we will derive an equation involving η based on this property.

Define the delta function δ_v as $\langle \delta_v, Y \rangle_{L^2} = (v, Y(x_0))$, here $(\cdot, \cdot)$ is the inner product on $(\text{Ad}P)_{x_0}$. Thus, for $\eta \in \tilde{H}_A^s$, since $\langle \delta_v, X \rangle_{L^2} = (v, X(x_0)) = 0$, there is some v in the fiber $(\text{Ad}P)_{x_0}$ for which η solves the distributional equation

$$d_A^*(\square_A^1)^s \eta = \delta_v, \quad (1.22)$$

To obtain a more useful and direct form of equation (1.22), we separately consider the cases where A is reducible and irreducible.

Suppose that A is irreducible. The Laplacian $\Delta_A^0 = d_A^* d_A$ on $\Omega^0(\text{Ad}P)$ is invertible. Let $G_A(x, y)$ denote the Green function of Δ_A^0 , i.e. the section over $M \times M$ with values in $(\text{Ad}P)_x^* \otimes (\text{Ad}P)_y$ which satisfies $\Delta_A^0 G_A(x, y)v = \delta_v(y)$, for distinct x, y , $G_A(x, y)$ is a linear map from $(\text{Ad}P)_x$ to $(\text{Ad}P)_y$. Fixing $x = x_0$, for each $v \in (\text{Ad}P)_{x_0}$, put $G_A^v(y) = G_A(x_0, y)v$, then the section $G_A^v(y)$ of $\text{Ad}P$ is smooth for $y \neq x_0$, is singular at $y = x_0$, and satisfies $\Delta_A^0 G_A^v(y) = \delta_v(y)$. Hence (1.22) can be written as $d_A^*((\square_A^1)^s \eta - d_A G_A^v) = 0$.

Suppose that A is reducible. The Laplacian Δ_A^0 is not invertible. In this case we can solve (1.22) only for certain $v \in (\text{Ad}P)_{x_0}$. Indeed, if v satisfies (1.22) and $\Phi \in \ker(\Delta_A^0) = \ker(d_A)$, then

$$(v, \Phi(x_0)) = \langle \delta_v, \Phi \rangle_{L^2} = \langle d_A^*(\square_A^1)^s \eta, \Phi \rangle_{L^2} = \langle (\square_A^1)^s \eta, d_A \Phi \rangle_{L^2} = 0.$$

Hence all the v 's which satisfies (1.22) lies in the orthogonal complement to the subspace $K_{x_0} = \text{span}\{\Phi(x_0) \mid \Phi \in \ker(d_A)\} \subset (\text{Ad}P)_{x_0}$. Conversely, when $v \perp K_{x_0}$, as revealed by the spectral decomposition of self-dual elliptic operator, Δ_A^0 is invertible on the L^2 -orthogonal complement of its kernel, and this inverse is given by convolution with the Green function defined by

$$G_A(x, y) = \sum_{\lambda_i > 0} \lambda_i^{-1} (\phi_i(x), \cdot) \phi_i(y), \quad (1.23)$$

where $\{\phi_i \in \Omega^0(\text{Ad}P)\}$ is an L^2 -orthonormal basis of eigenfunctions with eigenvalues $\{\lambda_i\}$, and this modified Green function (1.23) is an projection onto $(\ker \Delta_A^0)^\perp$. Fixing $x = x_0$, we obtain sections $G_A^v(y)$ of $\text{Ad}P$ as above; these are again singular at $y = x_0$ and smooth elsewhere, and satisfy (denote $\{\Phi_j\}$ is an L^2 -orthonormal basis of $\ker(d_A)$)

$$\begin{aligned} \Delta_A^0 G_A^v(y) &= \Delta_A^0 (G_A(x_0, y)v) \\ &= \Delta_A^0 \left(\sum_{\lambda_i > 0} \lambda_i^{-1} (\phi_i(x_0), v) \phi_i(y) \right) \\ &= \sum_{\lambda_i > 0} (\phi_i(x_0), v) \phi_i(y) \\ &= \sum_{\lambda_i > 0} \langle \delta_v, \phi_i \rangle_{L^2} \phi_i(y) \\ &= \delta_v(y) - \sum_j \langle \delta_v, \Phi_j \rangle_{L^2} \Phi_j(y) \\ &= \delta_v(y) - \sum_j (\Phi_j(x_0), v) \Phi_j(y), \end{aligned}$$

we used the completeness of L^2 -space, i.e. for any section $f(y)$,

$$f(y) = \sum_{\lambda_i > 0} \langle \phi_i, f \rangle_{L^2} \phi_i(y) + \sum_j \langle \Phi_j, f \rangle_{L^2} \Phi_j(y).$$

In particular, when $v \perp K_{x_0}$ we again have $\Delta_A^0 G_A^v = \delta_v$. Thus for all connections A , the equation (1.22) defining the slice $\tilde{H}_A$ is equivalent to

$$d_A^*((\square_A^1)^s \eta - d_A G_A^v) = 0 \quad (1.24)$$

for some $v \perp K_{x_0}$. Whenever we write G_A^v we assume $v \perp K_{x_0}$.

Now suppose A is self-dual. Then $\square_A^0 d_A^* = d_A^* \square_A^1$ and $d_A \square_A^0 = \square_A^1 d_A$, so multiplying (1.24) by $(\square_A^0)^{-s}$ gives

$$d_A^*(\eta - d_A(\square_A^0)^{-s} G_A^v) = 0. \quad (1.25)$$

Hence when A is self-dual, Hodge Decomposition Theorem reveals

$$\tilde{H}_A^s = \text{span}\{H_A, d_A(\square_A^0)^{-s} G_A^v\} = H_A \oplus \text{span}\{d_A(\square_A^0)^{-s} G_A^v\} \quad (1.26)$$

and this splitting is L_s^2 -orthogonal. Thus for self-dual A the L_s^2 -horizontal slice for the action of $\mathcal{G}_{s+1}^0$ on $\mathcal{A}_s$ is spanned by the Δ_A^0 -harmonic forms (which are smooth) and the sections $d_A(\square_A^0)^{-s} G_A^v$.

Remark 1. *If we let A vary through self-dual connections, then as we pass through a reducible connection A_0 some of the sections G_A^v vary discontinuously; furthermore, $\dim(H_A \cap \ker d_A^-)$ jumps up by $\dim(\ker(d_{A_0}))$ and*

$$\dim(\text{span}\{d_A(\square_A^0)^{-s} G_A^v\})$$

jumps down. Nevertheless, one can show that $\tilde{\mathcal{H}}_A^s = \tilde{H}_A^s \cap \ker(d_A^-)$ varies smoothly.

Before we state the theorem on smooth Hilbert structure and metric structure of orbit space $\tilde{\mathcal{B}}_s = \mathcal{A}_s / \mathcal{G}_{s+1}^0$, we need the following theorem arise from functional analysis:

Lemma 1.6. *For every $A \in \mathcal{A}_s$, $s > 1$, there is a constant C s.t. $\forall X \in \mathfrak{g}_{s+1}^0$,*

$$\|X\|_{L_{s+1}^2} \leq C \|d_A X\|_{L_s^2}.$$

Proof. Assume that the inequality does not hold. Then, there exists a sequence $\{X_n\} \subset \mathfrak{g}_{s+1}^0$ such that:

$$\|X_n\|_{L_{s+1}^2} = 1, \quad \|d_A X_n\|_{L_s^2} \rightarrow 0.$$

Since the embedding $L_{s+1}^2 \hookrightarrow C^0$ is compact, there exists a subsequence $\{X_{n_k}\}$ that converges in the C^0 -norm to some $X_\infty \in C^0$. Furthermore, since L_{s+1}^2 is a Hilbert space, which means L_{s+1}^2 is reflexive, thus Banach-Alaoglu Theorem implies that the unit ball in L_{s+1}^2 is compact in the weak*-topology, which means $X_{n_k} \rightharpoonup X'_\infty$. Since X_{n_k} converges to X_∞ strongly in C^0 space, we deduce $X_\infty = X'_\infty$, i.e.

$$X_{n_k} \rightharpoonup X_\infty \quad \text{in } L_{s+1}^2.$$

For continuous linear operator $d_A: L_{s+1}^2 \rightarrow L_s^2$ we have $d_A X_{n_k} \rightharpoonup d_A X_\infty$ in L_s^2 , combine this fact with $\|d_A X_{n_k}\|_{L_s^2} \rightarrow 0$ we deduce $d_A X_\infty = 0$. Note that for every $A \in \mathcal{A}_s$, $s > 1$, there is a Poincaré inequality for $\mathfrak{g}_{s+1}^0$: there is a constant C such that $\|X\|_{L^2} \leq C\|d_A X\|_{L^2}$ for every $X \in \mathfrak{g}_{s+1}^0$ (cf. Morrey's book, "Multiple integrals in the calculus of variations" Theorem 3.6.5, using the facts that the embedding $L_{s+1}^2 \hookrightarrow C^0$ is compact and that $d_A X = 0$ implies $X = 0$ as above), we obtain $X_\infty = 0$, this contradicts the fact $\|X_n\|_{L_{s+1}^2} = 1$.

□

Now we can prove the following

Theorem 1.7. *For each $s > 1$, $\tilde{\mathcal{B}}_s = \mathcal{A}_s/\mathcal{G}_{s+1}^0$ is a smooth Hausdorff manifold and $\mathcal{A}_s \rightarrow \tilde{\mathcal{B}}_s$ is a smooth fibration of Hilbert manifolds. There is a natural (strong) L_s^2 metric on $\tilde{\mathcal{B}}_s$ which makes this fibration a Riemannian submersion.*

Proof. In the context of this theorem, the central idea is to employ the slice theorem in order to establish a local diffeomorphism that connects $\mathcal{A}_s$ and $\tilde{\mathcal{B}}_s$. Here are concrete steps.

Step1. Smooth Hausdorff Hilbert structure on $\tilde{\mathcal{B}}_s$.

We begin by showing that (1.24) defines a slice for the action of $\mathcal{G}_{s+1}^0$. Fix $A \in \mathcal{A}_s$ and consider the affine subspace $S_A = A + \tilde{H}_A^s \subset \mathcal{A}_s$. The group action gives a map

$$\Phi: \mathcal{G}_{s+1}^0 \times S_A \rightarrow \mathcal{A}_s,$$

which is smooth for $s > 1$ (cf. Uhlenbeck's paper, "Connections with L^p bounds on curvature", Lemma 1.2). Identifying $T_A S_A$ with $\tilde{H}_A^s$, the differential of Φ at (Id, A) is

$$\begin{aligned} D\Phi: \mathfrak{g}_{s+1}^0 \times \tilde{H}_A^s &\rightarrow T_A \mathcal{A}_s, \\ (X, \eta) &\mapsto d_A X + \eta. \end{aligned}$$

Now we check $D\Phi|_{(\text{Id}, A)}$ is isomorphism.

- $D\Phi|_{(\text{Id}, A)}$ is injective.

Suppose that $d_A X + \eta = 0$, and put $\eta = \eta_1 + C d_A(\square_A^0)^{-s} G_A^v$, then

$$\begin{aligned} \langle d_A X, d_A X \rangle_s &= -\langle d_A X, \eta \rangle_s \\ &= -\langle d_A X, (\square_A^1)^s \eta \rangle_{L^2} \\ &= -\langle X, d_A^* (\square_A^1)^s \eta \rangle_{L^2} \\ &= -\langle X, d_A^* d_A G_A^v \rangle_{L^2} \\ &= -\langle X, \delta_v \rangle_{L^2} \\ &= 0 \end{aligned}$$

Therefore $d_A X = 0$. In particular $|X| = \text{const} = 0$ since $X(x_0) = 0$. Hence $X = 0$, $\eta = 0$, and $D\Phi|_{(\text{Id}, A)}$ is injective.

- $D\Phi|_{(\text{Id}, A)}$ is surjective.

lemma 1.6 implies that $d_A \mathfrak{g}_{s+1}^0 \subset L_s^2(E)$ is closed. More precisely, suppose $d_A X_n \rightarrow \eta$ in L_s^2 , since

$$\|X_n - X_m\|_{L_{s+1}^2} \leq C \|dX_n - dX_m\|_{L_s^2},$$

We deduce that $\{X_n\}$ is a Cauchy sequence in L_{s+1}^2 , then there exists $X_n \rightarrow X_\infty$ in L_{s+1}^2 . Notice the fact: $d_A: L_{s+1}^2 \rightarrow L_s^2$ is a closed operator, hence $d_A X_\infty = \eta$, which implies $\eta \in \text{Im } d_A$ and $\text{Im } d_A$ is closed in $\Omega_s^1(\text{Ad}P)$. This fact induces Hodge decomposition:

$$\Omega_s^1(\text{Ad}P) = \text{Im } d_A \oplus \tilde{H}_A^s,$$

Therefore, every element of $T_A \mathcal{A}_s = \Omega_s^1(\text{Ad}P)$ lies in the image of $D\Phi|_{(\text{Id}, A)}$.

We have proved $D\Phi|_{(\text{Id}, A)}$ is an isomorphism, so by the inverse function theorem Φ is a local diffeomorphism of neighborhoods $\mathcal{O}_1 \times \mathcal{O}_2 \rightarrow \mathcal{O}_s$. Currently, we have illuminated the affine subspace $S_A = A + \tilde{H}_A^s \subset \mathcal{A}_s$ satisfying transversality:

$$T_A S_A \oplus \text{Im } d_A = T_A \mathcal{A}_s.$$

For the purpose of constructing a slice, we still need to reveal the local uniqueness, i.e. the slice intersects each gauge orbit at most once. Specifically, assume there exist $g \neq \text{Id} \in \mathcal{G}_{s+1}^0$ and $A_1, A_2 \in S_A \cap \mathcal{O}_2$ such that $g \cdot A_1 = A_2$, we need to show that $g = \text{Id}$. Actually, an argument of Atiyah in his paper, "Self-duality in four-dimensional Riemannian Geometry" suffices to conclude the proof.

To reveal $\tilde{\mathcal{B}}_s = \mathcal{A}_s / \mathcal{G}_{s+1}^0$ is a smooth Hausdorff manifold, we need to verify

- Local smooth Hilbert structure.

The local diffeomorphism $\Phi: \mathcal{O}_1 \times \mathcal{O}_2 \rightarrow \mathcal{O}_s$ induces local charts on $\tilde{\mathcal{B}}_s$ via projection:

$$\pi: \mathcal{A}_s \rightarrow \tilde{\mathcal{B}}_s, \quad \pi(A + \eta) = [A + \eta].$$

For each slice neighborhood $\mathcal{O}_2 \subset S_A$, the map $\pi|_{\mathcal{O}_2}: \mathcal{O}_2 \rightarrow \tilde{\mathcal{B}}_s$ is a homeomorphism onto its image. The transition maps between different slices are smooth because the gauge action Φ is smooth. Thus, $\tilde{\mathcal{B}}_s$ inherits a smooth Hilbert manifold structure from $\mathcal{A}_s$.

- Hausdorff property.

We prove the Hausdorff property by contradiction. Suppose $\tilde{\mathcal{B}}_s$ is not Hausdorff, then there exist two distinct gauge orbits $[A_1] \neq [A_2]$ in $\tilde{\mathcal{B}}_s$, and sequences $\{A_n\} \subset \mathcal{A}_s, \{g_n\} \subset \mathcal{G}_{s+1}^0$, such that $A_n \in S_{A_1} = A_1 + \mathcal{O}_2$ (where S_{A_1} is a slice neighborhood)

and $g_n \cdot A_n \rightarrow A_2$ in $\mathcal{A}_s$ with $A_2 \in S_{A_2} = A_2 + \mathcal{O}'_2$. Since $A_n \in S_{A_1}$, we write $A_n = A_1 + \eta_n, \eta_n \in \tilde{H}_{A_1}^s \cap \mathcal{O}_2$.

For the local diffeomorphism $\Phi: \mathcal{O}_1 \times \mathcal{O}_2 \rightarrow \mathcal{O}_s$, denote $\Phi^{-1}(A'_n) = (g_n, A_n)$. Since $A'_n = \Phi(g_n, A_n)$ converges to A_2 , by the continuity of Φ^{-1} , we have (g_n, A_n) converges to $\Phi^{-1}(A_2)$. Moreover, notice that $\Phi(\text{Id}, A_2) = A_2$, which means the pre-image of A_2 only contains one element (Id, A_2) , hence g_n must converge to Id , and A_n must converge to A_2 . However, $A_n \in S_{A_1}$, which implies $A_2 \in \overline{S}_{A_1}$. This contradicts the transversality of slices unless $[A_1] = [A_2]$. Hence, $\tilde{\mathcal{B}}_s$ is Hausdorff.

Step2: $\mathcal{A}_s \rightarrow \tilde{\mathcal{B}}_s$ as a smooth fibration.

The projection $\pi: \mathcal{A}_s \rightarrow \tilde{\mathcal{B}}_s$ is a principal $\mathcal{G}_{s+1}^0$ -bundle. Locally, $\mathcal{A}_s$ is diffeomorphic to $\mathcal{G}_{s+1}^0 \times \tilde{\mathcal{B}}_s$, with the gauge group acting freely and properly on the fibers. The smoothness of π follows from the smoothness of the slice map Φ and the implicit function theorem.

Step3: Riemannian submersion structure.

The tangent space $T_A \mathcal{A}_s$ splits orthogonally as:

$$T_A \mathcal{A}_s = \text{Im}(d_A) \oplus \tilde{H}_A^s,$$

where $\text{Im}(d_A)$ is the vertical subspace (tangent to gauge orbits) and $\tilde{H}_A^s$ is the horizontal subspace. The L_s^2 -metric on $\mathcal{A}_s$ restricts to $\tilde{H}_A^s$, defining a metric on $T_{[A]} \tilde{\mathcal{B}}_s$. This metric is gauge-invariant (so the result is independent of the choice of A and is well defined) and makes π a Riemannian submersion, as the differential $D\pi$ preserves the horizontal metric:

$$\langle \eta_1, \eta_2 \rangle_{L_s^2} = \langle D\pi(\eta_1), D\pi(\eta_2) \rangle_{L_s^2}, \quad \forall \eta_1, \eta_2 \in \tilde{H}_A^s.$$

□

As noted in the proof above, the L_s^2 metric on $\mathcal{A}_s$ is gauge-invariant, so the residual action of $\mathcal{G}/\mathcal{G}_0 \cong G/Z$ on $\tilde{\mathcal{B}}$ is isometric. This action is free on $\tilde{\mathcal{B}}^* \subset \tilde{\mathcal{B}}$ as we verified in lemma 1.4. Thus we have a commutative diagram of Riemannian submersions,

$$\begin{array}{ccc} \mathcal{A}_s^* & & \\ \pi' \downarrow & \searrow \pi'_0 & \\ \mathcal{B}_s^* & \xleftarrow{\pi'_1} & \tilde{\mathcal{B}}_s^* \end{array} \quad (1.27)$$

where all spaces have their L_s^2 metrics; moreover π'_1 is a principal G/Z -bundle. We can now restrict this diagram to the self-dual connections. Let $\mathcal{A}' = \{A \in \mathcal{A} \mid \ker(d_A^-)^* = \{0\}\}$. An application of the implicit function theorem shows that $\mathcal{SD}'_s = \mathcal{SD}_s \cap \mathcal{A}'$ is a smooth Hilbert manifold for $s > 1$ (cf. Freed and Uhlenbeck's book, "Instantons and four-manifolds", section 3), namely

Theorem 1.8. *Let $\mathcal{A}' = \{A \in \mathcal{A} \mid \ker(d_A^-)^* = \{0\}\}$, then $\mathcal{SD}'_s = \mathcal{SD}_s \cap \mathcal{A}'$ is a smooth Hilbert manifold for $s > 1$. Moreover, $T_A(\mathcal{SD}') = \ker(d_A^-)$ holds.*

The proof for theorem 1.8 mainly base on the following Implicit Function Theorem:

Theorem 1.9 (Implicit Function Theorem (Banach Space Version)). *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces, and let $\Phi : \mathcal{X} \rightarrow \mathcal{Y}$ be a smooth map. Suppose that at a point $A_0 \in \mathcal{X}$, the following conditions are satisfied:*

1. $\Phi(A_0) = 0$.
2. $D\Phi_{A_0}$ is **Surjective**.
3. **Closed Splitting of the Kernel:** *there exists closed complementary space Z of the kernel of the differential operator $D\Phi_{A_0}$ such that the space $\mathcal{X}$ can be decomposed as:*

$$\mathcal{X} = \ker D\Phi_{A_0} \oplus Z.$$

Then there exists a neighborhood of A_0 in $\mathcal{X}$ where the equation $\Phi(A) = 0$ implicitly defines a smooth manifold. Specifically, the solution space $\Phi^{-1}(0)$ near A_0 is a smooth Hilbert submanifold of $\mathcal{X}$, and the tangent space to this manifold at A_0 is given by

$$T_{A_0}\Phi^{-1}(0) = \ker D\Phi_{A_0}. \quad (1.28)$$

Thus, there exists a smooth map $g : \ker D\Phi_{A_0} \rightarrow \mathcal{X}$ that locally parametrizes the solutions of $\Phi(A) = 0$.

proof for theorem 1.8. Now we use above “IFT” to show $\mathcal{SD}'_s = \mathcal{SD}_s \cap \mathcal{A}'$ is a smooth Hilbert manifold and $T_A(\mathcal{SD}') = \ker(d_A^-)$ holds.

Define the curvature map:

$$\Psi : \mathcal{A}_s \rightarrow \Omega_-^2(\text{Ad } P), \quad \Psi(A) = F_A^-,$$

where $F_A^- = p_-(F_A)$ is the anti-self-dual part of the curvature. The derivative of Φ at a connection $A \in \mathcal{SD}_s$ can be computed, $\forall \eta \in T_A\mathcal{A}_s \cong \Omega^1(\text{Ad } P)$,

$$\begin{aligned} D\Psi_A(\eta) &= \left. \frac{d}{dt} \right|_{t=0} p_- F_{A+t\eta} \\ &= \left. \frac{d}{dt} \right|_{t=0} p_- (d(A+t\eta) + (A+t\eta) \wedge (A+t\eta)) \\ &= p_- (d\eta + A \wedge \eta + \eta \wedge A) \\ &= p_- d_A \eta. \end{aligned}$$

Consider the fundamental elliptic complex associated with A :

$$0 \rightarrow \Omega^0(\text{Ad} P) \xrightarrow{d_A} \Omega^1(\text{Ad} P) \xrightarrow{d_A^-} \Omega_-^2(\text{Ad} P) \rightarrow 0,$$

By elliptic regularity of Laplacian $\Delta_A = d_A^* d_A + (d_A^-)^* d_A^-$, we deduce $\ker(\Delta_A)$ is finite-dimensional, and $d_A^- : \Omega^1(\text{Ad} P) \rightarrow \Omega_-^2(\text{Ad} P)$ is a Fredholm operator. For $A \in \mathcal{A}' = \{A \mid \ker(d_A^-)^* = 0\}$, the operator d_A^- has trivial cokernel:

$$\text{coker}(d_A^-) = \ker(d_A^-)^* = \ker(d_A^* p_+) = 0.$$

Thus, $D\Phi_A$ is surjective.

Since $d\Psi_A$ at $A \in \mathcal{A}'$ is a surjective Fredholm operator, by the implicit function theorem 1.9, there exists an open neighborhood $U_A \subset \mathcal{A}$ such that $F^{-1}(0) \cap U_A \subset \mathcal{A}$ is a smooth Hilbert submanifold with tangent space $\ker(d\Psi_A)$. Furthermore, since $\ker(d_A^-)^* = \{0\}$ in $\mathcal{A}'$, this condition remains open in $\mathcal{A}'$, and thus $\mathcal{SD}'_s = F^{-1}(0) \cap \mathcal{A}'$ is locally a submanifold. More specifically, for each $A \in \mathcal{SD}'_s$, the implicit function theorem provides a diffeomorphism in a neighborhood U_A :

$$\Phi_A : U_A \rightarrow V_A \subset \ker(d_A^-) \times \mathbb{R}^n,$$

where Φ_A maps the solution set $F^{-1}(0) \cap U_A$ to $\ker(d_A^-) \times \{0\}$. If two neighborhoods U_A and $U_{A'}$ have a non-empty intersection, then the transition function:

$$\Phi_{A'} \circ \Phi_A^{-1} : \ker(d_A^-) \rightarrow \ker(d_{A'}^-)$$

is smooth, because both Ψ and the parameterization by the implicit function theorem are smooth mappings. When $s > 1 + \frac{\dim M}{2}$, the Sobolev embedding $H^s \hookrightarrow C^k$ ensures that the higher-order derivatives of the coordinate transformations exist and are continuous, thus establishing the overall smooth structure.

According to the implicit function theorem, the tangent space at A of the solution space $\mathcal{SD}'_s$ is given by the kernel of the derivative $\ker(D\Psi_A)$. Here $D\Psi_A = d_A^-$, thus $T_A(\mathcal{SD}') = \ker(d_A^-)$.

□

Writing $\mathcal{M}'_s = \mathcal{SD}'_s / \mathcal{G}_{s+1}$ and $\widetilde{\mathcal{M}}'_s = \mathcal{SD}'_s / \mathcal{G}_{s+1}^0$, we have a commutative diagram

$$\begin{array}{ccc} \mathcal{SD}'_s & & \\ \downarrow \pi & \searrow \pi_0 & \\ \mathcal{M}' & \xleftarrow{\pi_1} & \widetilde{\mathcal{M}}'_s \end{array} \quad (1.29)$$

where π_0 is a Riemannian submersion, and π, π_1 are submersions over the subspace $\mathcal{M}'^* = \mathcal{M}' \cap \mathcal{B}^*$.

Since $T_A(\mathcal{SD}') = \ker(d_A^-)$, we obtain slices for π and π_0 by intersecting H_A^s and $\widetilde{H}_A^s$ with $\ker(d_A^-)$. Denoting these restricted slices by $\mathcal{H}_A^s$ and $\widetilde{\mathcal{H}}_A^s$, we thus have identifications

$$\begin{aligned} T_{[A]}\mathcal{M}'^* &= \mathcal{H}_A^s \stackrel{\text{def}}{=} H_A^s \cap \ker(d_A^-), \\ T_{[A]}\widetilde{\mathcal{M}}'_s &= \widetilde{\mathcal{H}}_A^s \stackrel{\text{def}}{=} \widetilde{H}_A^s \cap \ker(d_A^-). \end{aligned} \quad (1.30)$$

These will be used frequently in later sections. We now state the main theorem of this section. It describes the natural Riemannian metrics on the moduli space $\widetilde{\mathcal{M}}'$.

Theorem 1.10. *1) The differentiable structure on $\widetilde{\mathcal{M}}'_s$ is independent of $s > 1$. This based moduli space $\widetilde{\mathcal{M}}' = \widetilde{\mathcal{M}}'_s$ is thus a well-defined smooth finite-dimensional manifold.*

2) $\widetilde{\mathcal{M}}'$ carries an analytic family of smooth metrics $\{g_s \mid s > 1\}$ induced by the L_s^2 norms. The action of G/Z on $\widetilde{\mathcal{M}}'$ is isometric with respect to each g_s .

3) For each $s > 1$ there is a commutative diagram (1.29) where $\widetilde{\mathcal{M}}'_s = \widetilde{\mathcal{M}}'$. The map π_0 is a Riemannian submersion with respect to the L_s^2 -induced metrics and, over $\mathcal{M}'^$, π is a Riemannian submersion with respect to both the L^2 - and L_s^2 -induced metrics. For each $s > 1$, π_1 is a Riemannian submersion from $(\widetilde{\mathcal{M}}'^*, g_s)$ to $(\mathcal{M}'^*, g)$.*

sketch of the proof. (a) **Smooth structure and diffeomorphisms:** For any $1 \leq s \leq s'$, the inclusions $\mathcal{A}_{s'} \hookrightarrow \mathcal{A}_s$ and $\mathcal{G}_{s'+1}^0 \hookrightarrow \mathcal{G}_{s+1}^0$ induce a smooth inclusion map $\widetilde{\mathcal{B}}_{s'} \rightarrow \widetilde{\mathcal{B}}_s$ which restricts to a map $\iota: \widetilde{\mathcal{M}}'_{s'} \rightarrow \widetilde{\mathcal{M}}'_s$. Since every self-dual connection is gauge-equivalent to a smooth one, ι is a bijection. The slice theorem, combined with the regular value theorem, gives smooth maps $\Phi_{s'}, \Phi_s$ from neighborhoods of $[A]$ in $\mathcal{SD}'_{s'}$ (respectively, U of $\iota([A])$ in $\mathcal{SD}'_s$) to $\widetilde{\mathcal{H}}_A^{s'}$ (respectively $\widetilde{\mathcal{H}}_A^s$), defining smooth local charts. To prove that the smooth structures on $\widetilde{\mathcal{M}}'_{s'}$ and $\widetilde{\mathcal{M}}'_s$ are equivalent, it suffices to show that the overlap map $\widetilde{\mathcal{H}}_A^{s'} \rightarrow \widetilde{\mathcal{H}}_A^s$ is locally a diffeomorphism. Using the fact that $\Psi_{s'}^{-1}$ is an affine map, we conclude that the overlap map is locally a diffeomorphism.

(b) **L_s^2 -metrics:** The L_s^2 metric on $\widetilde{\mathcal{M}}'_s$ (or $\widetilde{\mathcal{M}}'^*$) is simply the restriction of the L_s^2 metric on $\mathcal{A}_s$ to $\widetilde{\mathcal{H}}_A^s$. Since the action of $\mathcal{G}_{s+1}$ on $\mathcal{A}_s$ is isometric, it preserves the distribution $\widetilde{\mathcal{H}}_A^s$ and induces a G/Z -isometric action on $\widetilde{\mathcal{M}}'_s$.

(c) **Riemannian submersions:** The maps π_0 and π are Riemannian submersions of the L_s^2 metrics. The map π is an L^2 -Riemannian submersion, and we've verified that π_1 is also a Riemannian submersion. $\square$

Chapter 2 Geometry of Yang-Mills Moduli Space for Definite Manifolds,

I: Geometry of the Collar

This chapter provides a detailed analysis of Part II of the paper “The Geometry of Yang-Mills Moduli Space for Definite Manifolds” by D. Groisser and T.H. Parker. The focus is on understanding the Riemannian geometry of the “collar” end of the Yang-Mills moduli space $\mathcal{M}$ for $k = 1$ instantons on a compact, simply-connected 4-manifold M with a positive-definite intersection form. The main goal is to detail the proof of Theorem II in this paper, which establishes that the natural L^2 metric on the collar is asymptotically a product metric.

The moduli space $\mathcal{M}$ is a 5-dimensional manifold, except for a finite number of conical singularities. Away from a compact set, $\mathcal{M}$ has a “collar” end, which is diffeomorphic to $(0, \lambda_0) \times M$ for some small $\lambda_0 > 0$. The parameter $\lambda \in (0, \lambda_0)$ is the **scale** of an instanton, and $p \in M$ is its **center**. Points in the collar, $[A] \in \mathcal{M}$, correspond to instantons whose curvature $|F_A|^2$ is highly concentrated around a point $p \in M$ in a region of size λ .

Definition 2.1 (Collar Map, Donaldson). *For a sufficiently small λ_0 , there exists a diffeomorphism, called the collar map:*

$$\Psi : \mathcal{M}_{\lambda_0} \rightarrow (0, \lambda_0) \times M, \quad (2.1)$$

where $\mathcal{M}_{\lambda_0} = \{[A] \in \mathcal{M} \mid \lambda([A]) \leq \lambda_0\}$, and $\Psi([A]) = (\lambda([A]), p([A]))$.

Main Goal of This Chapter To prove the following theorem, which describes the L^2 metric g on $\mathcal{M}$ in the coordinates provided by Ψ .

Theorem 2.2 (Asymptotic Metric on the Collar). *Let h be the product metric $4\pi^2(2d\lambda^2 + g_M)$ on $\mathbb{R} \times M$, where g_M is the metric on M . Then the L^2 metric g on $\mathcal{M}$ is asymptotically equal to the pulled-back metric Ψ^*h as $\lambda \rightarrow 0$. More precisely, for any $\varepsilon > 0$, there exists a $\lambda_0 > 0$ such that for any $[A] \in \mathcal{M}_{\lambda_0}$ and any tangent vector $W \in T_{[A]}\mathcal{M}$:*

$$(1 - \varepsilon)\Psi^*h(W, W) \leq g(W, W) \leq (1 + \varepsilon)\Psi^*h(W, W). \quad (2.2)$$

The proof is structured in three main steps,

- Step 1: Construct an explicit, computable **approximate tangent space** $\mathcal{T}_A$ for the true tangent space $T_A\mathcal{M}$.
- Step 2: Compute the **differential of the collar map**, $d\Psi$.
- Step 3: Combine these results to prove theorem 2.2.

1 Step 1: Construct the Approximate Tangent Space

The true tangent space $T_A\mathcal{M}$ at a connection A is the space of harmonic 1-forms $\mathcal{H}_A = \{\omega \in \Omega^1(\text{Ad}P) \mid d_A\omega = 0, d_A^*\omega = 0\}$. These are defined by global PDE's and are hard to compute. The strategy is to construct an explicit, local approximation.

Definition 2.3 (Approximate Tangent Space). *Let $[[A]] \in \mathcal{M}_{\lambda_0}$ with center p and scale λ . Fix a function $b \in C_0^\infty(\mathbb{R})$ with $0 \leq b(t) \leq 1$, $b(t) = 1$ for $t \in [0, 1]$, $b(t) = 0$ for $t \geq 2$, and $b'(t) \leq 0$ for all t , define a cut-off function $\beta := b(r/r_0)$. In normal coordinates at p , define vector fields representing variations in center and scale:*

- $Y_{\mathbf{a}} = \text{grad}(\beta(r)f_{\mathbf{a}})$, where $f_{\mathbf{a}} = \mathbf{a}_i x^i$ for $\mathbf{a} \in T_p M$. This is a variation of the center.
- $X = \text{grad}(\beta(r)\frac{r^2}{2})$. This corresponds to a variation of the scale.

The **approximate tangent space** $\mathcal{T}_A$ is defined as the span of 1-forms obtained by contracting these vector fields with the curvature 2-form F_A :

$$\mathcal{T}_A = \text{span}\{\iota_X F_A, \iota_{Y_{\mathbf{a}}} F_A \mid \mathbf{a} \in T_p M\}. \quad (2.3)$$

The core of this section is to show that $\mathcal{T}_A$ is a good approximation of $T_A\mathcal{M}$.

Proposition 2.4. *The forms in $\mathcal{T}_A$ are "approximately harmonic". Specifically, the terms $\iota_X F_A$ and $\iota_{Y_{\mathbf{a}}} F_A$ have the following operational properties regarding the exterior derivative*

$$d_A^*(\iota_X F_A) = d_A^*(\iota_{Y_{\mathbf{a}}} F_A) = 0 \quad (2.4)$$

$$\|d_A^-(\iota_X F_A)\|_{L^2}^2 \leq C\lambda^{4-2\delta} \quad (2.5)$$

$$\|d_A^-(\iota_{Y_{\mathbf{a}}} F_A)\|_{L^2}^2 \leq C|\mathbf{a}|^2\lambda^2. \quad (2.6)$$

Proof. The proof of this proposition relies on certain facts, which we shall introduce in advance and state without proof,

Let $Z = \text{grad}(\phi)$ be a gradient vector field and A a self-dual connection, then

$$d_A^*(\iota_Z F_A) = 0, \quad (2.7)$$

$$d_A^-(\iota_Z F_A) = \sum_{j,k} (H^0 \phi)_{jk} \theta^j \wedge (\iota_{e_k} F_A). \quad (2.8)$$

Here ι_Z denotes contraction with Z , $\{e_j\}$ is any local orthonormal basis of TM , $\{\theta^j\}$ is the dual coframe, and $H^0 \phi$ is the traceless part of the covariant Hessian $H\phi$ of ϕ .

Now back to the proof of proposition 2.2. Notice that X and $Y_{\mathbf{a}}$ have the form $Z = \text{grad}(\beta f)$, where f is either $\frac{1}{2}r^2$ or $f_{\mathbf{a}}$. Hence (2.7) immediately gives statement (2.4). For (2.5) and (2.6), we apply Lemma (2.8) to $\phi = \beta f$. We have

$$H(\beta f) = \nabla d(\beta f) = (H\beta)f + (d\beta \otimes df + df \otimes d\beta) + \beta(Hf),$$

so (2.8) gives the pointwise bound

$$|d_A^-(\iota_Z F_A)| \leq C(|f||H^0 \beta| + 2|d\beta||df| + \beta|H^0 f|)|F_A|. \quad (2.9)$$

Now

$$\begin{aligned} d\beta &= r_0^{-1} b'(r/r_0) dr, \\ \nabla d\beta &= r_0^{-2} b''(r/r_0) dr \otimes dr + r_0^{-1} b'(r/r_0) \nabla dr, \\ \nabla dr &= r^{-1} (x^i \nabla dx^i + dx^i \otimes dx^i - dr \otimes dr). \end{aligned}$$

Since $\nabla dx^i = -\Gamma_{jk}^i dx^j \otimes dx^k$ and $r^{-1} \leq C$ on Ω , we have

$$|d\beta| \leq C\chi_\Omega, \quad |H^0 \beta| \leq |H\beta| \leq C\chi_\Omega. \quad (2.10)$$

Substituting into (2.9) we find

$$|d_A^-(\iota_Z F_A)| \leq C\chi_\Omega(|f| + |df|)|F_A| + c\chi_B |H^0 f||F_A|, \quad (2.11)$$

whence

$$\|d_A^-(\iota_Z F_A)\|_2^2 \leq C \int_\Omega (|f| + |df|)^2 |F_A|^2 + C \int_B |H^0 f|^2 |F_A|^2. \quad (2.12)$$

We apply this in two cases.

- **Case 1.** Taking $f = \frac{1}{2}r^2$ (i.e. $Z = X$), we wish to show that $\chi_B |H^0 f| \leq C\chi_B r^2$ for $f = \frac{1}{2}r^2$, where $H^0 f$ is the traceless part of the Hessian of f .

Let us work in Riemannian normal coordinates (x^i) centered at the point p (the center of the ball B). In these coordinates, the metric tensor g satisfies

$$\begin{aligned} g_{ij}(p) &= \delta_{ij}, \quad \partial_k g_{ij}(p) = 0, \\ g_{ij}(x) &= \delta_{ij} + O(r^2). \end{aligned}$$

This relation formally implies that for $x \in B$ (where B is a sufficiently small ball), there exists a constant C_1 such that $|g(x) - g_{\text{flat}}| \leq C_1 r^2$, where $g_{\text{flat}} = dx^i \otimes dx^i$ is the Euclidean metric.

The Hessian of $f = \frac{1}{2}r^2 = \frac{1}{2} \sum_k (x^k)^2$ has components

$$H_{ij} = \nabla_i(\nabla_j f) = \partial_i(\nabla_j f) - \Gamma_{ij}^k \nabla_k f = \partial_i(x^j) - \Gamma_{ij}^k x^k = \delta_{ij} - \Gamma_{ij}^k x^k.$$

Since $\Gamma_{ij}^k(p) = 0$, we have $\Gamma_{ij}^k(x) = O(r)$. Therefore, the term $\Gamma_{ij}^k x^k = O(r) \cdot O(r) = O(r^2)$. This gives us a key estimate for the Hessian tensor itself,

$$H_{ij} = \delta_{ij} + O(r^2), \quad \text{or equivalently,} \quad |Hf - g_{\text{flat}}| \leq C_2 r^2. \quad (2.13)$$

The traceless part of the Hessian is defined as $H^0 f = Hf - \frac{1}{4}(\text{Tr}_g Hf)g$. Let us first estimate the trace term, $\text{Tr}_g Hf$,

$$\text{Tr}_g Hf = g^{ij} H_{ij} = (\delta^{ij} + O(r^2))(\delta_{ij} + O(r^2)) = 4 + O(r^2).$$

Now, we can estimate $H^0 f$,

$$\begin{aligned} |H^0 f| &= \left| (Hf - g_{\text{flat}}) + (g_{\text{flat}} - g) + \left(1 - \frac{1}{4}\text{Tr}_g Hf\right)g \right| \\ &\leq C_2 r^2 + C_1 r^2 + C_3 r^2. \end{aligned}$$

Multiplying by the characteristic function χ_B to localize the estimate, we arrive at the desired result:

$$\chi_B |H^0 f| \leq C \chi_B r^2.$$

- **Case 2.** Taking $f = a_i x^i$ (i.e., $Z = Y_{\mathbf{a}}$), we seek to establish the estimate $\chi_B |H^0 f| \leq \chi_B |\nabla df| \leq \chi_B \cdot C |\mathbf{a}| r$ for the linear function $f = a_i x^i$.

The inequality $|H^0 f| \leq |\nabla df|$ is fundamental. The Hessian $H = \nabla df$ admits an orthogonal decomposition $H = H^0 + \frac{1}{4}(\text{Tr}_g H)g$.

To bound $|\nabla df|$, we still work in Riemannian normal coordinates centered at p . Let $f(x) = a_i x^i$, where $\mathbf{a} = (a_i)$ is a constant vector. The components of its gradient are $\nabla_j f = \partial_j f = a_j$. The Hessian $H_{ij} = \nabla_i(\nabla_j f)$ is then given by

$$H_{ij} = \nabla_i a_j = \partial_i a_j - \Gamma_{ij}^k a_k = -\Gamma_{ij}^k a_k.$$

Taking the pointwise norm of the Hessian, we get

$$|\nabla df|(x) = |H(x)| \leq C_1 |\Gamma(x)| |\mathbf{a}| \leq C r |\mathbf{a}|.$$

Combining these two steps, we have $|H^0 f| \leq |\nabla df| \leq C |\mathbf{a}| r$. Localizing this to the ball B with the characteristic function χ_B yields the desired estimate:

$$\chi_B |H^0 f| \leq \chi_B |\nabla df| \leq C \chi_B |\mathbf{a}| r.$$

In each case we can substitute the appropriate bounds into (2.12) and estimate the integrals using a fact:

A Fact on Curvature Estimate. There exists a constant C such that, given $\delta > 0$, there exist $r_0 > 0$ and $\lambda_0 = \lambda_0(\delta) > 0$ such that the curvature of each $[A] \in M_{\lambda_0}$ satisfies

$$|F_A|(q) \leq C\lambda^{2-\delta}/r^{4-\delta}, \quad (2.14)$$

whenever $r = \text{dist}(q, p([A])) \leq r_0$ (here $\lambda = \lambda([A])$).

First, we have

$$\int_{\Omega} |F_A|^2 \leq C\lambda^{4-2\delta} \int_{\Omega} r^{2\delta-8} \leq C\lambda^{4-2\delta}. \quad (2.15)$$

Next, we bound the integrals over B by writing $B = B_{\lambda} \cup (B - B_{\lambda})$, where $B_{\lambda} = \{r \leq \lambda\}$, and using estimate (2.14) on the annulus $B - B_{\lambda}$. Thus for $p \in \mathbb{Z}$,

$$\begin{aligned} \int_B r^p |F_A|^2 &\leq C\lambda^p \int_{B_{\lambda}} |F_A|^2 + C\lambda^{4-2\delta} \int_{B-B_{\lambda}} r^{p+2\delta-8} \\ &\leq C\lambda^p + C\lambda^{4-2\delta} \text{sgn}(p+2\delta-4) \left(r^{p+2\delta-4} \Big|_{r=\lambda}^{r=r_0} \right). \end{aligned} \quad (2.16)$$

since $\int_{B_{\lambda}} |F_A|^2 \leq \|F_A\|_2^2 \leq 8\pi^2$. In Case 1, (2.12), (2.15) and (2.16) yield

$$\|d_A^-(\iota_Z F_A)\|_2^2 \leq C\lambda^{4-2\delta} + C(\lambda^4 + \lambda^{4-2\delta}),$$

and statement (2.5) of the proposition follows. Similarly, in Case 2, (2.12), (2.15) and (2.16) plus the fact that $4 - 2\delta > 2$, give statement (2.6). □

We now present the key computational result in the form of the following proposition. These results beautifully connect the intrinsic L^2 geometry of the approximating 1-forms to the extrinsic coordinate parameters a and λ .

Proposition 2.5. *For each $\varepsilon > 0$, $\exists \lambda_0 > 0$ such that for all A with $\lambda(A) = \lambda \leq \lambda_0$,*

$$\|\iota_{Y_{\mathbf{a}}} F_A\|_{L^2}^2 = 4\pi^2 |\mathbf{a}|^2 (1 + O(\lambda^2)), \quad (2.17)$$

$$\|\iota_X F_A\|_{L^2}^2 = 8\pi^2 \lambda^2 (1 + O(\varepsilon)), \quad (2.18)$$

$$|\langle \iota_X F_A, \iota_{Y_{\mathbf{a}}} F_A \rangle| \leq C\varepsilon \|\iota_X F_A\| \|\iota_{Y_{\mathbf{a}}} F_A\|. \quad (2.19)$$

where, in (2.18), we mean $|\|\iota_X F_A\|_2^2 - 8\pi^2 \lambda^2| \leq \varepsilon \lambda^2$.

Proof. Prior to the proof of this proposition, we state without proof the following results that will be employed.

Lemma 2.6. *For any two vector fields Z and Z' , and any two self-dual bundle-valued 2-forms F and F' ,*

$$(\iota_Z F, \iota_{Z'} F') + (\iota_{Z'} F, \iota_Z F') = (Z, Z')(F, F') \quad (2.20)$$

pointwise. In particular,

$$(\iota_Z F, \iota_{Z'} F) = \frac{1}{2}(Z, Z')|F|^2. \quad (2.21)$$

Lemma 2.7. *Let $h : [0, \infty) \rightarrow \mathbb{R}$ be piecewise continuous with compact support. For each multi-index $I = (i_1, \dots, i_n)$ let x^I be the corresponding monomial on $\mathbb{R}^4$ and let $S \subset \mathbb{R}^4$ be the unit sphere. Define constants $K_I = \int_S x^I$ and $c_{m,n} = \int_0^\infty 48h(t)t^{n-m+3}(1+t^2)^{-4} dt$ (if $n - m + 3 \leq -1$ we assume $\text{supp}(h) \subset (0, \infty)$). Now let $\{x^i\}$ be normal coordinates on M at $p(A)$. Then as $\lambda = \lambda(A) \rightarrow 0$*

$$\lim_{\lambda \rightarrow 0} \left(\lambda^{m-n} \int_M h(r/\lambda) x^I r^{-m} |F_A|^2 \right) = c_{m,n} K_I, \quad (2.22)$$

uniformly in $[A] \in \mathcal{M}$. In particular, when $n = 1, 2$ this holds with $K_i = 0$ and $K_{ij} = \frac{1}{2}\pi^2 \delta_{ij}$.

Now back to the proof of this proposition. Let ϕ, ψ be functions equal to either $\frac{1}{2}r^2$ or $a_i x^i$, so $Z = \text{grad}(\beta\phi)$ and $Z' = \text{grad}(\beta\psi)$ are either X or $Y = Y_{\mathbf{a}}$. By the product rule and (2.10),

$$\begin{aligned} |(Z, Z') - \beta^2(d\phi, d\psi)| &= |(d\beta, \phi\psi d\beta + \beta\phi d\psi + \beta\psi d\phi)| \\ &\leq C(|\psi\phi| + |d(\phi\psi)|)\chi_\Omega. \end{aligned}$$

Now let $p = \deg(\phi\psi)$ (so $p = 2, 3$, or 4) and let $m = \text{homogeneity of } |\mathbf{a}| \text{ in } \phi\psi$ (so $m = 0, 1$, or 2). Then $|\phi\psi| \leq c|\mathbf{a}|^m r^p$ and $|d(\phi\psi)| \leq c|\mathbf{a}|^m r^{p-1}$. Multiplying the inequality above by $\frac{1}{2}|F|^2$ and using (2.21) gives

$$\begin{aligned} \left| (\iota_Z F, \iota_{Z'} F) - \frac{1}{2}\beta^2(d\phi, d\psi)|F|^2 \right| &\leq C|F|^2 |\mathbf{a}|^m (r^p + r^{p-1})\chi_\Omega \\ &\leq C|F|^2 |\mathbf{a}|^m \chi_\Omega, \end{aligned}$$

since r and r^{-1} are bounded on Ω . Integrating over M and using equation (2.15) yields

$$\langle \iota_Z F_A, \iota_{Z'} F_A \rangle = \frac{1}{2} \int \beta^2 |F_A|^2 (d\phi, d\psi) + O(|\mathbf{a}|^m \lambda^{4-2\delta}). \quad (2.23)$$

We will apply this in each of the three cases.

Case 1. Taking $\phi = \psi = a_i x^i$, we have $p = m = 2$, $d\phi = d\psi = a_i dx^i$, and hence $(d\phi, d\psi) = g^{ij} a_i a_j = |\mathbf{a}|^2(1 + O(r^2))$. Then (2.23) gives

$$\begin{aligned} \langle \iota_Y F, \iota_Y F \rangle &= \frac{1}{2} |\mathbf{a}|^2 \left(\int \beta^2 |F|^2 (1 + O(r^2)) + O(\lambda^{4-2\delta}) \right) \\ &= \frac{1}{2} |\mathbf{a}|^2 \left(\int \beta^2 |F|^2 + O(\lambda^2) \right), \end{aligned}$$

where we have used (2.16) and the fact that $4 - 2\delta > 2$. To compute the magnitude of $\iota_Y F$, we shall again require the following fact.

A Fact on Energy Estimate. Let $\Omega \subset M$ be the complement of the ball $B(p, \sqrt{r_0\lambda})$, where $p = p([A])$ and $\lambda = \lambda([A])$. Then there are constants λ_0 and C such that each $[A] \in \mathcal{M}_{\lambda_0}$ satisfies

$$\int_{\Omega} |F_A|^2 \leq C\lambda^2. \quad (2.24)$$

Now write $\beta^2 = 1 - (1 - \beta^2)$. Note that $\|F\|_2^2 = 8\pi^2$, and $1 - \beta^2$ has support in $M - B(p, r_0)$. When λ is sufficiently small we can then apply (2.24) to obtain

$$\|\iota_Y F\|_2^2 - 4\pi^2|\mathbf{a}|^2 \leq C|\mathbf{a}|^2\lambda^2.$$

Case 2. Next take $\phi = \psi = \frac{1}{2}r^2$, so $p = 4$, $m = 0$, $d\phi = d\psi = x^i dx^i$, and $(d\phi, d\psi) = g^{ij}x^i x^j = r^2$ (in normal coordinates, $g^{ij}x^j = x^i = g_{ij}x^j$). Choose $N \geq 1$ such that $N^{2\delta-2} < \varepsilon$ and let λ_0 be small enough that $N\lambda_0 < r_0$. Then $\beta \equiv 1$ on $\{r \leq N\lambda\}$ for $\lambda \leq \lambda_0$, so (2.23) gives

$$\begin{aligned} \|\iota_X F\|_2^2 &= \frac{1}{2} \int \beta^2 r^2 |F|^2 + O(\lambda^{4-2\delta}) \\ &= \frac{1}{2} \int_{r \leq N\lambda} r^2 |F|^2 + \frac{1}{2} \int_{r > N\lambda} \beta^2 r^2 |F|^2 + O(\lambda^{4-2\delta}). \end{aligned} \quad (2.25)$$

Now apply lemma 2.7 in the case $h = \text{characteristic function of } [0, N]$, $n = 0$, $m = -2$, to find

$$\begin{aligned} \lim_{\lambda \rightarrow 0} \left(\lambda^{-2} \int_{r \leq N\lambda} r^2 |F|^2 \right) &= \text{Vol}(S^3) \cdot \int_0^N 48t^5(1+t^2)^{-4} dt \\ &= 16\pi^2 \left(1 - \frac{3N^4 + 3N^2 + 1}{(N^2 + 1)^3} \right), \end{aligned}$$

by a direct calculation. But $(3N^4 + 3N^2 + 1)(N^2 + 1)^{-3} \leq 7N^{-2} \leq 7\varepsilon$, so, by taking λ_0 sufficiently small,

$$\left| \frac{1}{2} \int_{r \leq N\lambda} r^2 |F|^2 - 8\pi^2\lambda^2 \right| \leq 8\varepsilon\lambda^2. \quad (2.26)$$

On the other hand, estimate (2.14) reads

$$\int_{r > N\lambda} \beta^2 r^2 |F|^2 \leq C\lambda^{4-2\delta}(N\lambda)^{2\delta-2} \leq C\varepsilon\lambda^2. \quad (2.27)$$

Combine (2.25), (2.26) and (2.27), we deduce

$$\|\iota_X F_A\|_{L^2}^2 = 8\pi^2\lambda^2(1 + O(\varepsilon)).$$

Case 3. Take $\phi = \frac{1}{2}r^2$ and $\psi = a_i x^i$, so $p = 3$, $m = 1$, and $(d\phi, d\psi) = \psi$. (2.23) gives

$$\langle \iota_X F, \iota_Y F \rangle = \frac{1}{2} \int \beta^2 \psi |F|^2 + O(|\mathbf{a}| \lambda^{4-2\delta}).$$

Choosing $N \geq 1$ such that $N^{2\delta-3} < \varepsilon$, and choose λ_0 small enough that $N\lambda_0 \leq r_0$. Then, if $\lambda \leq \lambda_0$, we have

$$\int \beta^2 \psi |F|^2 = \int_{r \leq N\lambda} \psi |F|^2 + \int_{r > N\lambda} \beta^2 \psi |F|^2.$$

For the outer integral, we apply the argument used in (2.27), obtaining

$$\left| \int_{r > N\lambda} \beta^2 \psi |F|^2 \right| \leq c|\mathbf{a}| \lambda \varepsilon.$$

For the inner integral, lemma 2.7 implies

$$\lim_{\lambda \rightarrow 0} \left(\lambda^{-1} \int_{r \leq N\lambda} x^i |F|^2 \right) = 0,$$

Therefore, for λ sufficiently small, we have $|\int \psi |F|^2| \leq \varepsilon |\mathbf{a}| \lambda$. Putting all this together,

$$|\langle \iota_X F, \iota_Y F \rangle| \leq c|\mathbf{a}| \lambda \varepsilon$$

for λ sufficiently small. By expansions (2.17) and (2.18), $\|\iota_Y F\|$ and $\|\iota_X F\|$ are commensurate with $|\mathbf{a}|$ and λ , respectively. Hence estimate (2.19) follows. $\square$

Proposition 2.8. *There exist constants $C, \lambda_0 > 0$ such that if $Z = \text{grad}(\beta(\frac{1}{2}a_0 r^2 + f_{\mathbf{a}}))$ and $\lambda \leq \lambda_0$ then*

$$\|(1 - \pi_A) \iota_Z F_A\| \leq C(|a_0| \lambda^{2-\delta} + |\mathbf{a}| \lambda), \quad (2.28)$$

$$\|\pi_A \iota_Z F_A\| = \|\iota_Z F_A\| (1 + O(\lambda^{1-\delta})). \quad (2.29)$$

Proof. For any $\omega \in \Omega^1(\text{Ad}P)$, $(1 - \pi_A)\omega$ is the projection of ω onto the L^2 -orthogonal complement of the harmonic space. This can be expressed in terms of the Laplacians and Green operators of the fundamental elliptic complex, namely $(1 - \pi_A)\omega = d_A G_A^0 d_A^* \omega + (d_A^-)^* G_A^2 d_A^- \omega$. Taking $\omega = \iota_Z F$ we have $d_A^* \omega = 0$ by (2.4). Hence

$$\begin{aligned} \|(1 - \pi_A)\omega\|_2^2 &= \|(d_A^-)^* G_A^2 d_A^- \omega\|_2^2 \\ &= \langle G_A^2 d_A^- \omega, d_A^- (d_A^-)^* G_A^2 d_A^- \omega \rangle \\ &= \langle G_A^2 d_A^- \omega, d_A^- \omega \rangle. \end{aligned} \quad (2.30)$$

Let $L = d_A^- (d_A^-)^*$ be the Laplacian on $\Omega_-^2(\text{Ad}P)$, for which G_A^2 is the Green's operator. Let $\{\eta_k\}$ be an orthonormal eigenbasis for L with corresponding eigenvalues $\{\nu_k\}$, such

that $L\eta_k = \nu_k\eta_k$. We expand $\eta := d_A^-\omega$ in this basis as $\eta = \sum_k c_k\eta_k$, where $c_k = \langle \eta, \eta_k \rangle$. Applying the Green's operator, which acts as L^{-1} , gives $G_A^2\eta = \sum_k (c_k/\nu_k)\eta_k$. The inner product is then computed as

$$\langle G_A^2\eta, \eta \rangle = \left\langle \sum_k \frac{c_k}{\nu_k} \eta_k, \sum_j c_j \eta_j \right\rangle = \sum_k \frac{|c_k|^2}{\nu_k}.$$

It is a known result that there exists a constant $\mu > 0$ such that all non-zero eigenvalues of L satisfy $\nu_k \geq \mu$ (c.f. Proposition 18(ii) of Donaldson's paper, "*An application of gauge theory to four dimensional topology*"). This implies their reciprocals are bounded: $1/\nu_k \leq 1/\mu$. Applying this bound, we have

$$\langle G_A^2\eta, \eta \rangle \leq \frac{1}{\mu} \|\eta\|_2^2.$$

Substituting back $\eta = d_A^-\omega$, and thanks to proposition 2.4, we obtain (2.28).

Finally, we proof (2.29), and we break this down rigorously. Applying the triangle inequality yields

$$\|\pi_A\omega\| = \|\omega - (1 - \pi_A)\omega\| \leq \|\omega\| + \|(1 - \pi_A)\omega\|.$$

To compare the non-harmonic component $\|(1 - \pi_A)\iota_Z F\|$ with $\|\iota_Z F\|$, recall that $Z = a_0X + Y_a$, by proposition 2.5, we deduce

$$\|\iota_Z F\| \approx 2\pi\sqrt{2a_0^2\lambda^2 + |\mathbf{a}|^2}.$$

Now consider the ratio

$$\frac{\|(1 - \pi_A)\iota_Z F\|}{\|\iota_Z F\|} \lesssim \frac{|a_0|\lambda^{2-\delta} + |\mathbf{a}|\lambda}{\sqrt{2a_0^2\lambda^2 + |\mathbf{a}|^2}}.$$

We analyze two cases,

- If $|\mathbf{a}| \gtrsim |a_0|\lambda$, the denominator is $\sim |\mathbf{a}|$, and the numerator is $\sim |\mathbf{a}|\lambda$, so the ratio is $O(\lambda)$.
- If $|\mathbf{a}| \ll |a_0|\lambda$, the denominator is $\sim |a_0|\lambda$, and the numerator is $\sim |a_0|\lambda^{2-\delta}$, so the ratio is $O(\lambda^{1-\delta})$.

In both cases, since $\lambda \ll \lambda^{1-\delta}$ for small λ , the ratio is bounded by $C\lambda^{1-\delta}$. Hence,

$$\|(1 - \pi_A)\iota_Z F\| \leq C\lambda^{1-\delta} \|\iota_Z F\|.$$

Combining both steps, we obtain

$$\begin{aligned} \|\pi_A\iota_Z F\| &\leq \|\iota_Z F\| + \|(1 - \pi_A)\iota_Z F\| \\ &\leq \|\iota_Z F\| + C\lambda^{1-\delta} \|\iota_Z F\| \\ &= (1 + C\lambda^{1-\delta}) \|\iota_Z F\|. \end{aligned}$$

This shows that $\pi_A\iota_Z F$ is asymptotically commensurate with $\iota_Z F$ as $\lambda \rightarrow 0$, which completes the argument. □

2 Step 2: Compute the Differential of the Collar Map

The overarching goal of Part II of the paper is to understand the Riemannian geometry of the collar of the moduli space $\mathcal{M}$, which consists of instantons with small scale λ . This region is diffeomorphic to $(0, \lambda_0) \times M$ via the *collar map* Ψ . To analyze the L^2 metric g on $\mathcal{M}$, the authors compare it to a simple product metric on $(0, \lambda_0) \times M$. This comparison requires understanding how tangent vectors on $\mathcal{M}$ are transformed by Ψ . Therefore, the central task of this step is to perform a detailed calculation of the differential of the collar map, Ψ_* . The resulting formulas, while technical, are the essential input for the subsequent discussion, where the main theorems about the geometry of the collar are proven.

This section relies heavily on the construction of the collar map Ψ by Donaldson, which associates to each small-scale instanton $[A]$ a scale $\lambda([A]) \in \mathbb{R}^+$ and a center $p([A]) \in M$.

Definition 2.9 (Collar Map Characterization). *Let $b \in C_0^\infty([0, \infty))$ be the fixed smooth cutoff function used before. For a connection A , define the function*

$$R_A(s, x) := \int_M b(\text{dist}(x, y)/s) |F_A(y)|^2 d\text{Vol}_g(y); \quad (2.31)$$

$$\lambda(A) := K^{-1} \inf\{s \mid R_A(s, x) = 4\pi^2 \text{ for some } x\}. \quad (2.32)$$

For a sufficiently small-scale instanton $[A]$, its scale $\lambda = \lambda([A])$ and center $p = p([A])$ are uniquely defined by the following system of equations. Let $\bar{\lambda} = K\lambda$, where K is a normalization constant,

$$R_A(\bar{\lambda}(A), p(A)) = 4\pi^2 \quad (2.33)$$

$$\frac{\partial R_A}{\partial x^i}(\bar{\lambda}(A), p(A)) = 0 \quad (2.34)$$

where $\{x^i\}$ are normal coordinates around p . The **collar map** is then given by

$$\Psi : \mathcal{M}_{\lambda_0} \rightarrow (0, \lambda_0) \times M, \quad \Psi([A]) = (\lambda(A), p(A))$$

where $\mathcal{M}_{\lambda_0}$ is the collar, i.e., the set of instantons with scale $\lambda \leq \lambda_0$.

The main result of this section is a collection of propositions that together provide an asymptotic formula for $\Psi_* = (\lambda_*, p_*)$.

Firstly, The differential of the scale λ is

Proposition 2.10. *Let $A_t = A + t\eta + O(t^2)$ be a path of self-dual connections passing through A at $t = 0$, with tangent vector $[\eta] = \pi_*\eta \in T_{[A]}\mathcal{M}$. Let $\gamma(y) = b(\text{dist}(p, y)/\bar{\lambda})$. Then*

$$\bar{\lambda}_*[\eta] = -4\lambda \frac{\int (\eta, \iota_{\nabla\gamma} F_A)}{\int (\nabla\gamma, \nabla(r^2)) |F_A|^2}. \quad (2.35)$$

Furthermore, the denominator is uniformly bounded away from zero and infinity for small λ .

Proof. The proof proceeds by differentiating the defining equation (2.33), $R_{A_t}(\bar{\lambda}_t, p_t) = 4\pi^2$, with respect to t at $t = 0$. By the chain rule,

$$\frac{d}{dt} R_{A_t}(\bar{\lambda}, p) + \frac{\partial R_A}{\partial x^i} \frac{dp^i}{dt} + \frac{\partial R_A}{\partial s} \frac{d\bar{\lambda}_t}{dt} = 0.$$

The middle term vanishes by (2.34). Solving for $\frac{d\bar{\lambda}_t}{dt}$ gives

$$\bar{\lambda}_*[\eta] = \frac{d\bar{\lambda}_t}{dt} \Big|_{t=0} = - \frac{(d/dt) R_{A_t}(\bar{\lambda}, p) \Big|_{t=0}}{(\partial R_A / \partial s)(\bar{\lambda}, p)}.$$

The numerator is calculated by differentiating the integral definition of R_{A_t} ,

$$\begin{aligned} \frac{d}{dt} R_{A_t}(\bar{\lambda}, p) \Big|_{t=0} &= \frac{d}{dt} \Big|_{t=0} \left(\int_M \gamma(y) |F_{A_t}(y)|^2 d\text{Vol}_g(y) \right) \\ &= 2 \int_M \gamma \left(\frac{dF_{A_t}}{dt} \Big|_{t=0}, F_A \right) d\text{Vol}_g \\ &= 2 \int_M \gamma (d_A \eta, F_A) d\text{Vol}_g \\ &= 2 \int_M (\eta, d_A^*(\gamma F_A)) d\text{Vol}_g \\ &= 2 \int_M (\eta, \gamma d_A^* F_A - \iota_{\nabla\gamma} F_A) d\text{Vol}_g \\ &= -2 \int_M (\eta, \iota_{\nabla\gamma} F_A) d\text{Vol}_g. \end{aligned}$$

The denominator is calculated by differentiating R_A with respect to the scale variable s ,

$$\frac{\partial R_A}{\partial s}(\bar{\lambda}, p) = \int \frac{\partial}{\partial s} b\left(\frac{r}{s}\right) \Big|_{s=\bar{\lambda}} |F_A|^2 = -\frac{1}{2\bar{\lambda}} \int (\nabla\gamma, \nabla(r^2)) |F_A|^2$$

Substituting these into the expression for $\bar{\lambda}_*[\eta]$ and using $\bar{\lambda} = K\lambda$ yields the result. The bounds on the denominator follow from lemma 2.7. □

Calculating p_* is more involved as it requires differentiating with respect to the center point, which moves the base of the normal coordinate system. This requires a technical lemma on the derivatives of the distance function.

Lemma 2.11. *Let $\rho(p) = \text{dist}(y, p)$. In normal coordinates $\{x^i\}$ centered at p , we have:*

$$\begin{aligned} \frac{\partial \rho}{\partial x^i}(p) &= -y^i/r, \\ \left| \frac{\partial^2 \rho}{\partial x^i \partial x^j}(p) - \left(\frac{\delta_{ij}}{r} - \frac{y^i y^j}{r^3} \right) \right| &\leq Cr, \end{aligned}$$

where $y^i = x^i(y)$ and $r = \text{dist}(p, y)$.

Proof Sketch. This is a standard Riemannian geometry calculation using Jacobi fields. The variation vector field for moving the endpoint p of a geodesic from y is a Jacobi field. The first and second variation of arc-length formulas, applied to this Jacobi field, give the first and second derivatives of the distance function ρ . □

This lemma allows the authors to control the Hessian of R_A with respect to the center point coordinates.

Lemma 2.12. *Let $H_{ij} = (\partial^2 R_A / \partial x^i \partial x^j)(\bar{\lambda}, p)$ be the Hessian of R_A . There exists a scale $\lambda_0 > 0$ and a constant $C > 0$, such that for any $[A] \in \mathcal{M}$ with scale $\bar{\lambda}(A) < \lambda_0$, the scaled Hessian matrix $\bar{\lambda}^2 H_{ij}$ is uniformly negative definite:*

$$C^{-1}|\xi|^2 \leq -(\bar{\lambda}^2 H_{ij})\xi^i \xi^j \leq C|\xi|^2, \quad \forall \xi \in T_p M. \quad (2.36)$$

Moreover, the Hessian has the asymptotic expansion

$$\bar{\lambda}^2 H_{ij} = \int_M \bar{\lambda}^2 \gamma_{ij} |F_A|^2 d\text{Vol}_g + O(\bar{\lambda}^2), \quad (2.37)$$

where $\gamma(y) = b(\text{dist}(p, y)/\bar{\lambda})$ and $\gamma_{ij} = \partial_i \partial_j \gamma$. As $\bar{\lambda} \rightarrow 0$, $\bar{\lambda}^2 H_{ij}$ converges to $C_1 \delta_{ij}$ for a negative constant C_1 .

Proof. The proof proceeds by first decomposing the Hessian into a main term and an error term. We then analyze the asymptotic behavior of the main term and bound the error term. Finally, we combine these results using matrix perturbation theory to establish uniform invertibility.

Step 1: Decomposition of the Hessian Integrand

Let $\rho_y(p) := \text{dist}(y, p)$. The Hessian H_{ij} is obtained by differentiating $R_A(\bar{\lambda}, p)$ twice with respect to the coordinates $\{x^i\}$ of the point p . Let $\partial_i = \partial / \partial x^i$,

$$\begin{aligned} \bar{\lambda}^2 H_{ij} &= \bar{\lambda}^2 \int_M \frac{\partial^2}{\partial x^i \partial x^j} \left(b \left(\frac{\rho_y(p)}{\bar{\lambda}} \right) \right) |F_A|^2 d\text{Vol}_g \\ &= \int_M \left\{ b'' \left(\frac{\rho_y}{\bar{\lambda}} \right) (\partial_i \rho_y)(\partial_j \rho_y) + \bar{\lambda} b' \left(\frac{\rho_y}{\bar{\lambda}} \right) (\partial_i \partial_j \rho_y) \right\} |F_A|^2 d\text{Vol}_g. \end{aligned}$$

We compare this integrand with the components of the Hessian of $\gamma(y) = b(\text{dist}(p, y)/\bar{\lambda})$ with respect to the variable y in a normal coordinate system centered at p . Let $r(y) = \text{dist}(p, y)$ and let $\partial'_i = \partial/\partial y^i$. The term $\gamma_{ij} = \partial'_i \partial'_j \gamma$ is given by

$$\bar{\lambda}^2 \gamma_{ij}(y) = b''\left(\frac{r}{\bar{\lambda}}\right) (\partial'_i r)(\partial'_j r) + \bar{\lambda} b'\left(\frac{r}{\bar{\lambda}}\right) (\partial'_i \partial'_j r). \quad (2.38)$$

By symmetry of the distance function and using results on the derivatives of the distance function in normal coordinates (c.f. lemma 2.11), we have $\partial_i \rho_y(p) = -\partial'_i r(y)$. Also, the second derivatives differ by a term of order $O(r)$: $\partial_i \partial_j \rho_y(p) - \partial'_i \partial'_j r(y) = O(r)$, where the error arises from the curvature of the manifold M .

This implies the integrand of $\bar{\lambda}^2 H_{ij}$ can be written as

$$\left\{ \bar{\lambda}^2 \gamma_{ij}(y) + \bar{\lambda} \phi_{ij}(y) \right\},$$

where $\phi_{ij}(y) = b'(r/\bar{\lambda}) \cdot O(r)$. Since b' is supported on the annulus $\{\bar{\lambda} \leq r \leq 2\bar{\lambda}\}$, we have the estimate $|\phi_{ij}(y)| \leq C'r$ for a uniform constant C' , and the support of ϕ_{ij} is contained in $B_{2\bar{\lambda}}(p)$.

Integrating this decomposition gives

$$\bar{\lambda}^2 H_{ij} = \underbrace{\int_M \bar{\lambda}^2 \gamma_{ij} |F_A|^2 d\text{Vol}_g}_{=: M_{ij}} + \underbrace{\int_M \bar{\lambda} \phi_{ij} |F_A|^2 d\text{Vol}_g}_{=: E_{ij}}. \quad (2.39)$$

Step 2: Analysis of the Error Term E_{ij}

We bound the magnitude of each component of the error matrix E_{ij} :

$$\begin{aligned} |E_{ij}| &\leq \int_{B_{2\bar{\lambda}}(p)} \bar{\lambda} |\phi_{ij}| |F_A|^2 d\text{Vol}_g \leq \bar{\lambda} \int_{B_{2\bar{\lambda}}(p)} (C'r) |F_A|^2 d\text{Vol}_g \\ &\leq \bar{\lambda} (C' \cdot 2\bar{\lambda}) \int_M |F_A|^2 d\text{Vol}_g = 2C'\bar{\lambda}^2 (8\pi^2) = C_E \bar{\lambda}^2. \end{aligned}$$

The matrix norm of the error is thus of order $O(\bar{\lambda}^2)$, which establishes (2.37).

Step 3: Analysis of the Main Term M_{ij}

The main term M_{ij} is given by the integral of $\bar{\lambda}^2 \gamma_{ij} |F_A|^2$. From the expression for γ_{ij} , we split this integral into two parts,

$$M_{ij} = \int_M \left(b''\left(\frac{r}{\bar{\lambda}}\right) \frac{y^i y^j}{r^2} + \bar{\lambda} b'\left(\frac{r}{\bar{\lambda}}\right) \left(\frac{\delta_{ij}}{r} - \frac{y^i y^j}{r^3} \right) \right) |F_A|^2 d\text{Vol}_g.$$

We analyze the limit of this expression as $\bar{\lambda} \rightarrow 0$ by applying Lemma 2.7 to each component integral.

- **Term 1:** $\int_M b''(r/\bar{\lambda}) y^i y^j r^{-2} |F_A|^2 d\text{Vol}_g.$

We match this to the form $\bar{\lambda}^{m-n} \int_M h(r/\bar{\lambda}) x^I r^{-m} |F_A|^2$. Here, $h(t) = b''(t)$, the monomial is $x^I = y^i y^j$ (so $n = 2$), and the radial dependence is $r^{-m} = r^{-2}$ (so $m = 2$). The pre-factor is $\bar{\lambda}^{m-n} = \bar{\lambda}^0 = 1$, which matches. By Lemma 2.7, the limit is $c_{2,2} K_{ij}$.

$$\begin{aligned} \lim_{\bar{\lambda} \rightarrow 0} (\text{Term 1}) &= c_{2,2} K_{ij} \\ &= \left(\int_0^\infty 48 b''(t) t^{2-2+3} (1+t^2)^{-4} dt \right) \left(\frac{1}{2} \pi^2 \delta_{ij} \right) \\ &= 24\pi^2 \delta_{ij} \int_0^\infty b''(t) t^3 (1+t^2)^{-4} dt. \end{aligned}$$

- **Term 2:** $\int_M \bar{\lambda} b'(r/\bar{\lambda}) \delta_{ij} r^{-1} |F_A|^2 d\text{Vol}_g$.

We match this to $\delta_{ij} \bar{\lambda}^{m-n} \int_M h(r/\bar{\lambda}) x^I r^{-m} |F_A|^2$. Here, $h(t) = b'(t)$, the monomial is $x^I = 1$ (so $n = 0$), and $r^{-m} = r^{-1}$ (so $m = 1$). The pre-factor is $\bar{\lambda}^{m-n} = \bar{\lambda}^{1-0} = \bar{\lambda}$, which matches. The limit is $\delta_{ij} \cdot c_{1,0} \cdot K_\emptyset$, where $K_\emptyset = \text{Area}(S^3) = 2\pi^2$.

$$\begin{aligned} \lim_{\bar{\lambda} \rightarrow 0} (\text{Term 2}) &= \delta_{ij} \cdot c_{1,0} \cdot K_\emptyset \\ &= \delta_{ij} \left(\int_0^\infty 48 b'(t) t^{0-1+3} (1+t^2)^{-4} dt \right) (2\pi^2) \\ &= 96\pi^2 \delta_{ij} \int_0^\infty b'(t) t^2 (1+t^2)^{-4} dt. \end{aligned}$$

- **Term 3:** $-\int_M \bar{\lambda} b'(r/\bar{\lambda}) y^i y^j r^{-3} |F_A|^2 d\text{Vol}_g$.

We match this to $\bar{\lambda}^{m-n} \int_M h(r/\bar{\lambda}) x^I r^{-m} |F_A|^2$. Here, $h(t) = -b'(t)$, the monomial is $x^I = y^i y^j$ ($n = 2$), and $r^{-m} = r^{-3}$ ($m = 3$). The pre-factor is $\bar{\lambda}^{m-n} = \bar{\lambda}^{3-2} = \bar{\lambda}$, which matches. The limit is $c_{3,2} K_{ij}$.

$$\begin{aligned} \lim_{\bar{\lambda} \rightarrow 0} (\text{Term 3}) &= c_{3,2} K_{ij} \\ &= \left(\int_0^\infty 48 (-b'(t)) t^{2-3+3} (1+t^2)^{-4} dt \right) \left(\frac{1}{2} \pi^2 \delta_{ij} \right) \\ &= -24\pi^2 \delta_{ij} \int_0^\infty b'(t) t^2 (1+t^2)^{-4} dt. \end{aligned}$$

Combining these three limits, we find the limit of the main term M_{ij} ,

$$\begin{aligned} \lim_{\bar{\lambda} \rightarrow 0} M_{ij} &= 24\pi^2 \delta_{ij} \int_0^\infty (b''(t) t^3 + 4b'(t) t^2 - b'(t) t^2) (1+t^2)^{-4} dt \\ &= 24\pi^2 \delta_{ij} \int_0^\infty (b''(t) t^3 + 3b'(t) t^2) (1+t^2)^{-4} dt \\ &= 24\pi^2 \delta_{ij} \int_0^\infty (1+t^2)^{-4} \frac{d}{dt} (b'(t) t^3) dt. \end{aligned}$$

Using integration by parts, where the boundary terms vanish because b' has compact support in $(0, \infty)$, this becomes:

$$\begin{aligned} \lim_{\bar{\lambda} \rightarrow 0} M_{ij} &= -24\pi^2 \delta_{ij} \int_0^\infty b'(t) t^3 \frac{d}{dt} ((1+t^2)^{-4}) dt \\ &= -24\pi^2 \delta_{ij} \int_0^\infty b'(t) t^3 (-8t(1+t^2)^{-5}) dt \\ &= 192\pi^2 \delta_{ij} \int_0^\infty b'(t) t^4 (1+t^2)^{-5} dt. \end{aligned}$$

Since $b' \leq 0$ and is strictly negative on some interval, the integrand is non-positive and strictly negative somewhere. Therefore, the resulting integral is a strictly negative constant. Let us define this constant as C_1 ,

$$C_1 := 192\pi^2 \int_0^\infty b'(t) t^4 (1+t^2)^{-5} dt < 0.$$

This establishes that $\lim_{\bar{\lambda} \rightarrow 0} M_{ij} = C_1 \delta_{ij}$, completing the analysis of the main term.

Step 4: Uniform Invertibility

From Step 2 and 3, we have

$$\bar{\lambda}^2 H_{ij} = M_{ij} + E_{ij},$$

where $M_{ij} \rightarrow C_1 \delta_{ij}$ and $E_{ij} = O(\bar{\lambda}^2)$. This is a classic matrix perturbation problem. The matrix $C_1 \delta_{ij}$ is negative definite with eigenvalues equal to C_1 . By the convergence of M_{ij} , for any $\epsilon > 0$, there exists a $\lambda_1(\epsilon)$ such that for all $\bar{\lambda} < \lambda_1$, the eigenvalues of M_{ij} are contained in $[C_1 - \epsilon, C_1 + \epsilon]$. The error matrix norm satisfies $\|E\| \leq C'_E \bar{\lambda}^2$. To guarantee that the perturbed matrix $M + E$ remains negative definite, we must control the perturbation so that the least absolutely eigenvalue of the main part remains negative after perturbation. Let's choose $\epsilon = C_1/2$. Then for $\bar{\lambda} < \lambda_1(C_1/2)$, the largest eigenvalue of M_{ij} is less than $C_1/2$. We require

$$\|E\| \leq C'_E \bar{\lambda}^2 < -\frac{C_1}{4}.$$

This is satisfied if $\bar{\lambda} < \sqrt{-C_1/(4C'_E)}$. Let $\lambda_0 = \min(\lambda_1(C_1/2), \sqrt{-C_1/(4C'_E)})$. For any connection with scale $\bar{\lambda} < \lambda_0$, the largest eigenvalue of $\bar{\lambda}^2 H_{ij}$ is bounded above,

$$\mu_{\max}(\bar{\lambda}^2 H) \leq \mu_{\max}(M) - \|E\| \leq \frac{C_1}{2} + C'_E \bar{\lambda}^2 < \frac{C_1}{2} + C'_E \left(-\frac{C_1}{4C'_E}\right) = \frac{C_1}{4} < 0.$$

This implies there exists a constant C such that (2.36) holds, establishing uniform negative definiteness (and thus invertibility). $\square$

With these tools, the differential of the center point can be calculated.

Proposition 2.13. *There exist $C, \lambda_0 > 0$ such that if $[A] \in \mathcal{M}_{\lambda_0}$, and $A_t = A + t\pi_A \iota_Z F_A + O(t^2)$ is a path of self-dual connections, then for any vector field Z on M ,*

$$\left| (p_*)^j (\pi_A \iota_Z F_A) + (\bar{\lambda}^2 H)_{ij}^{-1} \left(\int \bar{\lambda}^2 (Z, \nabla \gamma_i) |F_A|^2 + m_i \bar{\lambda}_* (\pi_A \iota_Z F_A) \right) \right| \leq C \|(1 - \pi_A) \iota_Z F_A\|, \quad (2.40)$$

where

$$m_i = \bar{\lambda}^{-1} \int b''(r/\bar{\lambda}) y^i |F_A|^2. \quad (2.41)$$

Here $\bar{\lambda} = \bar{\lambda}(A)$, γ_i is given by

$$\gamma_i(y) = \bar{\lambda}^{-1} b'(r/\bar{\lambda}) \partial'_i r = \bar{\lambda}^{-1} b'(r/\bar{\lambda}) y^i / r,$$

$(p_*)^j$ are the components of the image of p_* in $T_{p(A)}M$ with respect to normal coordinates $\{x^i\}$ centered at $p(A)$, and H_{ij} is the Hessian $(\partial^2 R_A / \partial x^i \partial y^j)(\bar{\lambda}, p)$.

Proof. Write $\bar{\lambda}_t = \lambda(A_t)$, $p_t = p(A_t)$, $\bar{\lambda} = \bar{\lambda}(A)$, and $p = p(A)$. Differentiating (2.34) yields

$$\begin{aligned} 0 &= \frac{d}{dt} \Big|_{t=0} \left(\frac{\partial R_{A_t}}{\partial x^i}(\bar{\lambda}_t, p_t) \right) \\ &= \left(\frac{d}{dt} \left(\frac{\partial R_{A_t}}{\partial x^i}(\bar{\lambda}, p) \right) + \frac{\partial^2 R_A}{\partial x^i \partial x^j}(\bar{\lambda}, p) \frac{dp_t^j}{dt} + \frac{\partial^2 R_A}{\partial s \partial x^i}(\bar{\lambda}, p) \frac{d\bar{\lambda}_t}{dt} \right) \Big|_{t=0}. \end{aligned}$$

By lemma 2.12, H_{ij} is invertible, so we can solve this to obtain

$$\begin{aligned} \frac{dp_t^i}{dt} \Big|_{t=0} &= -(\bar{\lambda}^2 H)_{ij}^{-1} \left(\bar{\lambda}^2 \frac{d}{dt} \Big|_{t=0} \left(\frac{\partial R_{A_t}}{\partial x^i} \right) (\bar{\lambda}, p) \right. \\ &\quad \left. + \bar{\lambda}^2 \frac{\partial^2 R_A}{\partial s \partial x^i}(\bar{\lambda}, p) \bar{\lambda}_* (\pi_A \iota_Z F_A) \right). \end{aligned} \quad (2.42)$$

Now we provide a detailed derivation of the term $\bar{\lambda}^2 \frac{\partial^2 R_A}{\partial s \partial x^i}$ evaluated at $(s, p) = (\bar{\lambda}, p(A))$. Differentiating $R_A(s, p)$ with respect to the coordinate x^i of the center p . Using the chain rule and the result from lemma 2.11, we get

$$\begin{aligned} \frac{\partial R_A}{\partial x^i}(s, p) &= \int_M \frac{\partial}{\partial x^i} \left(b \left(\frac{r}{s} \right) \right) |F_A|^2 d\text{Vol}_g \\ &= - \int_M b' \left(\frac{r}{s} \right) \frac{y^i}{rs} |F_A|^2 d\text{Vol}_g. \end{aligned}$$

Next, we differentiate the expression above with respect to the scale parameter s . The integrand depends on s in two places, so we apply the product rule

$$\begin{aligned} \frac{\partial^2 R_A}{\partial s \partial x^i} &= - \int_M \frac{\partial}{\partial s} \left(b' \left(\frac{r}{s} \right) \frac{y^i}{rs} \right) |F_A|^2 d\text{Vol}_g \\ &= - \int_M \left(\left(\frac{\partial}{\partial s} b' \left(\frac{r}{s} \right) \right) \cdot \frac{y^i}{rs} + b' \left(\frac{r}{s} \right) \cdot \left(\frac{\partial}{\partial s} \frac{y^i}{rs} \right) \right) |F_A|^2 d\text{Vol}_g \\ &= \int_M \left(b'' \left(\frac{r}{s} \right) \frac{y^i}{s^3} + b' \left(\frac{r}{s} \right) \frac{y^i}{rs^2} \right) |F_A|^2 d\text{Vol}_g. \end{aligned}$$

Finally, we set $s = \bar{\lambda}$ and multiply by $\bar{\lambda}^2$

$$\bar{\lambda}^2 \frac{\partial^2 R_A}{\partial s \partial x^i} \Big|_{s=\bar{\lambda}} = \int_M \left(\frac{1}{\bar{\lambda}} b'' \left(\frac{r}{\bar{\lambda}} \right) y^i + \frac{1}{r} b' \left(\frac{r}{\bar{\lambda}} \right) y^i \right) |F_A|^2 d\text{Vol}_g.$$

We now decompose this expression into two terms.

$$\bar{\lambda}^2 \frac{\partial^2 R_A}{\partial s \partial x^i} = \underbrace{\int_M \left(\bar{\lambda}^{-1} b'' \left(\frac{r}{\bar{\lambda}} \right) y^i \right) |F_A|^2 d\text{Vol}_g}_{\text{Term A}} + \underbrace{\int_M \left(r^{-1} b' \left(\frac{r}{\bar{\lambda}} \right) y^i \right) |F_A|^2 d\text{Vol}_g}_{\text{Term B}}. \quad (2.43)$$

By comparing with the definition in (2.41), we see that Term A is exactly m_i . Therefore, we arrive at the decomposition

$$\bar{\lambda}^2 \frac{\partial^2 R_A}{\partial s \partial x^i} = m_i + \int_M r^{-1} b' \left(\frac{r}{\bar{\lambda}} \right) y^i |F_A|^2 d\text{Vol}_g. \quad (2.44)$$

It is straightforward to see that this last term vanishes by (2.31), (2.34) and lemma 2.11.

Next, we analyze the term $\frac{d}{dt}(\partial R_{A_t}/\partial x^i)$ in (2.42). Let $\eta = \pi_A \iota_Z F_A$. Since the path consists of self-dual connections, $A_t = A + t\eta + O(t^2)$, the curvature evolves as $F_{A_t} = F_A + td_A\eta + O(t^2)$. Notice that $\frac{\partial}{\partial x^i} b(r/\bar{\lambda}) = -\gamma_i(y)$, with this, the term we want to analyze is

$$\begin{aligned} \frac{d}{dt} \Big|_{t=0} \left(\frac{\partial R_{A_t}}{\partial x^i} \right) &= \frac{d}{dt} \Big|_{t=0} \int_M (-\gamma_i(y)) |F_{A_t}|^2 d\text{Vol}_g \\ &= - \int_M \gamma_i(y) \cdot \left(\frac{d}{dt} \Big|_{t=0} |F_{A_t}|^2 \right) d\text{Vol}_g \\ &= -2 \int_M \gamma_i(d_A \eta_t, F_A) d\text{Vol}_g. \end{aligned}$$

Using integration by parts (adjoint property of d_A and d_A^*), this becomes

$$\begin{aligned} -2 \int_M (\eta_t, d_A^*(\gamma_i F_A)) d\text{Vol}_g &= -2 \int_M (\eta_t, \gamma_i d_A^* F_A - \iota_{\nabla \gamma_i} F_A) d\text{Vol}_g \\ &= 2 \int_M (\eta_t, \iota_{\nabla \gamma_i} F_A) d\text{Vol}_g, \end{aligned}$$

where we used $d_A^* F_A = 0$ because A is a Yang-Mills connection. Now, we decompose $\eta_t = \pi_A \iota_Z F_A$ as $\iota_Z F_A - (1 - \pi_A) \iota_Z F_A$,

$$\begin{aligned} 2 \int_M (\iota_Z F_A - (1 - \pi_A) \iota_Z F_A, \iota_{\nabla \gamma_i} F_A) d\text{Vol}_g &= 2 \int_M (\iota_Z F_A, \iota_{\nabla \gamma_i} F_A) d\text{Vol}_g \\ &\quad - 2 \int_M ((1 - \pi_A) \iota_Z F_A, \iota_{\nabla \gamma_i} F_A) d\text{Vol}_g. \end{aligned}$$

Using lemma 2.6 (i.e., equation (2.21)), the first term is $\int_M (Z, \nabla \gamma_i) |F_A|^2$. For the second term, we apply the Cauchy-Schwarz inequality. This gives us the core estimate

$$\left| \frac{d}{dt} \left(\frac{\partial R_{A_t}}{\partial x^i} \right) \Big|_{t=0} - \int_M (Z, \nabla \gamma_i) |F_A|^2 \right| \leq 2 \|(1 - \pi_A) \iota_Z F_A\|_{L^2} \|\iota_{\nabla \gamma_i} F_A\|_{L^2}. \quad (2.45)$$

The function γ_i has support on the annulus $\{\bar{\lambda} \leq r \leq 2\bar{\lambda}\}$. A direct calculation shows that its gradient $\nabla\gamma_i$ is bounded on this support. Therefore, $\|\iota_{\nabla\gamma_i}F_A\|_{L^2}$ is bounded by a constant independent of λ and Z . Multiplying by $\bar{\lambda}^2$, we get the final estimate for the connection-dependent term:

$$\left| \bar{\lambda}^2 \frac{d}{dt} \left(\frac{\partial R_{A_t}}{\partial x^i} \right) \right|_{t=0} - \int_M \bar{\lambda}^2 (Z, \nabla\gamma_i) |F_A|^2 \leq C \|(1 - \pi_A)\iota_Z F_A\|.$$

Step 4: Assemble the Final Result.

We substitute our findings for the two main terms back into the rearranged equation (2.42). The term $\bar{\lambda}^2 \frac{\partial^2 R_A}{\partial s \partial x^i}$ is replaced by m_i using (2.44). The connection-dependent term $\bar{\lambda}^2 \frac{d}{dt}(\dots)$ is replaced by its approximation $\int \bar{\lambda}^2 (Z, \nabla\gamma_i) |F_A|^2$, introducing an error controlled by $\|(1 - \pi_A)\iota_Z F_A\|$.

Combining these results in equation (2.42) gives

$$\frac{dp_t^i}{dt} \Big|_{t=0} = -(\bar{\lambda}^2 H)_{ij}^{-1} \left(\int_M \bar{\lambda}^2 (Z, \nabla\gamma_i) |F_A|^2 + m_i \bar{\lambda}_* (\pi_A \iota_Z F_A) + O(\|(1 - \pi_A)\iota_Z F_A\|) \right).$$

Rearranging this equation to match the form in the proposition statement (2.40) concludes the proof. □

3 Step 3: Analyzing the Asymptotic Behavior of the Metric on the Collar

The primary goal of this section is to synthesize the analytical results from the preceding two sections to describe the large-scale geometry of the moduli space $\mathcal{M}$. Specifically, the authors aim to show that the natural L^2 Riemannian metric on the collar part of the moduli space is asymptotically a simple product metric. This result is crucial because it allows for a concrete geometric understanding of the compactification of the moduli space, transforming it from a purely topological concept into a geometric object (a singular manifold with boundary) whose properties like diameter and volume can be determined.

This section is the culmination of Part II of this paper, and relies heavily on the technical machinery developed in previous sections. Key prerequisites include

- **The Collar Map Ψ :** Donaldson's diffeomorphism $\Psi : \text{Collar of } \mathcal{M} \rightarrow (0, \lambda_0) \times M$, which assigns to each self-dual connection $[A]$ its scale $\lambda(A)$ and center $p(A)$. This section aims to understand the geometry of $\mathcal{M}$ by pulling back the metric via Ψ .
- **Approximate Tangent Space $\mathcal{T}_A$:** Constructed in Step 1, this is a finite-dimensional space of 1-forms, $\mathcal{T}_A = \text{Span}\{\iota_X F_A, \iota_{Y_A} F_A\}$, that approximates the true tangent space $T_{[A]}\mathcal{M}$ for connections $[A]$ deep in the collar (i.e., for $\lambda \rightarrow 0$).

- **Quantitative Estimates from Step 1 and Step 2:** The proofs in this section depend directly on the analytical estimates derived earlier,
 - **proposition 2.5 & proposition 2.8:** These provide crucial estimates for the L^2 norms of vectors in the approximate tangent space and show that the projection $\pi_A : \mathcal{T}_A \rightarrow T_{[A]}\mathcal{M}$ is close to the identity for small λ .
 - **proposition 2.10 & proposition 2.13:** These give formulas for the differential of the collar map, $\Psi_* = (\lambda_*, p_*)$, when applied to tangent vectors, linking the change in scale (λ) and center (p) to integrals involving the tangent vectors and the curvature form F_A .

To compare the true L^2 metric g on the moduli space with the product metric h on the model space, the authors need a “dictionary” between the tangent spaces of these two spaces. This dictionary is the differential of the collar map, Ψ_* . However, Ψ_* lacks a simple expression, making a direct comparison difficult. Therefore, we will construct an “approximate inverse” to Ψ_* .

Definition 2.14 (Approximate Inverse Map α_A). *Given a connection $[A]$ in the collar (with center p and scale λ) and a tangent vector $(\alpha_0, \mathbf{a})$ in the model space $\mathbb{R} \times T_p M$, define a map α_A as follows,*

1. *First, a vector field $Z_{(\alpha_0, \mathbf{a})} = \nabla(\beta(\frac{1}{2}\alpha_0 r^2 + f_{\mathbf{a}}))$ is defined, where β is a cut-off function, r is the distance function from the center p , and $f_{\mathbf{a}}$ is the linear function associated with the vector $\mathbf{a}$. This vector field combines the approximate vector fields for changing the scale (related to r^2) and changing the center (related to $f_{\mathbf{a}}$).*
2. *Then, the map $\alpha_A : \mathbb{R} \times T_p M \rightarrow \mathcal{T}_A$ is defined as*

$$\alpha_A(\alpha_0, \mathbf{a}) = -\iota_{Z_{(\alpha_0, \mathbf{a})}} F_A,$$

where $\iota_Z F_A$ is the interior product of the vector field Z with the curvature 2-form F_A , $\mathcal{T}_A$ is the approximate tangent space $\text{Span}\{\iota_X F_A, \iota_{Y_a} F_A\}$.

3. *Finally, by composing with the harmonic projection $\pi_A : \mathcal{T}_A \rightarrow T_{[A]}\mathcal{M}$, a map from the model tangent space to the true tangent space, $\pi_A \circ \alpha_A$, is obtained.*

This construction essentially says: if we want to move within the moduli space such that the rate of change of the instanton’s scale is α_0 and the rate of change of its center is $\mathbf{a}$, which tangent direction $W \in T_{[A]}\mathcal{M}$ should we follow? This definition provides an approximate answer: $W \approx \pi_A(-\iota_{Z_{(\alpha_0, \mathbf{a})}} F_A)$. The next crucial proposition is to prove just how good this approximation is.

Proposition 2.15 (Effectiveness of the Approximation). *The map constructed above, when composed with the differential of the collar map Ψ_* , is very close to the identity map in the limit $\lambda \rightarrow 0$. More precisely, the error is bounded as follows,*

$$|\lambda_*(\pi_A \alpha_A(\alpha_0, \mathbf{a})) - \alpha_0| \leq C(|\alpha_0| + |\mathbf{a}|)\lambda^{1-\delta}, \quad (2.46)$$

$$|p_*(\pi_A \alpha_A(\alpha_0, \mathbf{a})) - \mathbf{a}| \leq C(|\alpha_0| + |\mathbf{a}|)\lambda^{1-\delta}. \quad (2.47)$$

where λ_* and p_* are the components of Ψ_* onto $\mathbb{R}$ and M , respectively.

Proof. **Proof of (2.46)**

Write $F = F_A$ and take $\eta = \pi_A \alpha_A(a_0, \mathbf{a}) = -\iota_Z F + (1 - \pi_A)\iota_Z F$ in proposition 2.10. When λ is sufficiently small, $\beta \equiv 1$ on $\text{supp}(\gamma)$, so using lemma 2.6 we may replace $(\iota_Z F, \iota_{\nabla \gamma} F) = \frac{1}{2}(Z, \nabla \gamma) |F|^2$ by $\frac{1}{4}a_0(\nabla r^2, \nabla \gamma) |F|^2 + \frac{1}{2}a_i(\nabla x^i, \nabla \gamma) |F|^2$. Hence the numerator of (2.35) is

$$\begin{aligned} \int (\eta, \iota_{\nabla \gamma} F) &= -\frac{1}{4}a_0 \int (\nabla r^2, \nabla \gamma) |F|^2 - \frac{1}{2}a_i \int (dx^i, d\gamma) |F|^2 \\ &\quad + \int ((1 - \pi_A)\iota_Z F, \iota_{\nabla \gamma} F). \end{aligned} \quad (2.48)$$

Now $(dx^i, d\gamma)(y) = -\bar{\lambda}^{-1}b'(r/\bar{\lambda})g^{ij}y^j/r = -\bar{\lambda}^{-1}b'(r/\bar{\lambda})y^i/r + \text{error term}$, and the middle integral above vanishes as before in (2.44). Therefore, multiplying (2.48) by -4λ , dividing by the denominator of (2.35), and applying uniformly boundness of this denominator, we see that

$$\begin{aligned} |\lambda_*[\eta] - a_0| &\leq C\lambda \left| \langle (1 - \pi_A)\iota_Z F, \iota_{\nabla \gamma} F \rangle_2 \right| \\ &\leq C \|(1 - \pi_A)\iota_Z F\|_2 \|\lambda \iota_{\nabla \gamma} F\|_2 \\ &\leq C \|(1 - \pi_A)\iota_Z F\|_2 \\ &\leq C(|\alpha_0| + |\mathbf{a}|)\lambda^{1-\delta}. \end{aligned}$$

Where in the third inequality we used $|\nabla \gamma| \leq C\lambda^{-1}$, and in the forth inequality we used proposition 2.8.

Proof of (2.47)

We apply proposition 2.13. Again take λ small enough that $\beta \equiv 1$ on $\text{supp}(\gamma)$. Then, on $\text{supp}(\gamma)$,

$$\begin{aligned} (Z, \nabla \gamma_i) &= \left(d \left(\frac{1}{2} \lambda^{-1} a_0 r^2 + a_j y^j \right), d\gamma_i \right) \\ &= ((\lambda^{-1} a_0 y^j + a_j) dy^j, \gamma_{ik} dy^k) \\ &= (\lambda^{-1} a_0 y^j + a_j)(\gamma_{ij} + \gamma_{ik}(g^{jk} - \delta_{jk})). \end{aligned}$$

But $g^{jk} - \delta_{jk} = O(r^2)$, $|\gamma_{ik}| \leq C\lambda^{-2}$, and $\bar{\lambda} \leq r \leq 2\bar{\lambda}$ on $\text{supp}(\gamma)$, so

$$(Z, \nabla \gamma_i) = (\lambda^{-1} a_0 y^j + a_j) \gamma_{ij} + O(|a_0| + |\mathbf{a}|).$$

Thus, since $\int |F|^2 \leq C$,

$$\int \bar{\lambda}^2 (Z, \nabla \gamma_i) |F|^2 = a_j \int \bar{\lambda}^2 \gamma_{ij} |F|^2 + a_0 \lambda^{-1} \int y^j \bar{\lambda}^2 \gamma_{ij} |F|^2 + O((|a_0| + |\mathbf{a}|) \lambda^2).$$

By (2.37), the first term on the right is $\bar{\lambda}^2 H_{ij} a_j + O(|\mathbf{a}| \lambda^2)$. Since $\bar{\lambda}^2 y^j \gamma_{ij} = b''(r/\bar{\lambda}) y^i$, the second term on the right is $a_0 K m_i$, where m_i is as in (2.41), and K is as in (2.32).

Therefore

$$\int \bar{\lambda}^2 (Z, \nabla \gamma_i) |F|^2 + m_i \bar{\lambda}_* (\pi_A \iota_Z F_A) = \bar{\lambda}^2 H_{ij} a_j + K m_i (\lambda_* (\pi_A \iota_Z F_A) + a_0) + O((|a_0| + |\mathbf{a}|) \lambda^2). \quad (2.49)$$

Now $|m_i|$ is bounded (in fact lemma 2.7 shows that $m_i \rightarrow 0$ as $\lambda \rightarrow 0$), so we may use (2.46) to bound the second term on the right-hand side of (2.49). We can then combine (2.40), (2.49), proposition 2.8 and the fact that $\bar{\lambda}^2 H_{ij}$ is uniformly negative definite to obtain

$$|(p_*)^j (\pi_A \iota_Z F_A) - a_j| \leq C(|a_0| + |\mathbf{a}|) \lambda^{1-\delta},$$

and (2.47) then follows from the definition of α_A . □

With the core tool established, we are now ready to prove the most important theorem of this chapter.

Theorem 2.16 (Asymptotic Behavior of the Metric). *Let h_{cyl} be the product metric $h_{\text{cyl}} = 4\pi^2(2d\lambda^2 \oplus g_M)$ on the model space $(0, \lambda_0) \times M$, where g_M is the Riemannian metric on the 4-manifold M . Let Ψ be the collar map defined before. Then the L^2 metric g on the moduli space $\mathcal{M}$ is asymptotic to the pullback metric $\Psi^* h_{\text{cyl}}$ as $\lambda \rightarrow 0$. More precisely, for any $\varepsilon > 0$, there exists a $\lambda_0 > 0$ such that for any connection $[A] \in \mathcal{M}$ with $\lambda(A) < \lambda_0$ and any tangent vector $W \in T_{[A]}\mathcal{M}$, the following holds,*

$$(1 - \varepsilon)(\Psi^* h_{\text{cyl}})(W, W) \leq g(W, W) \leq (1 + \varepsilon)(\Psi^* h_{\text{cyl}})(W, W). \quad (2.50)$$

Proof. Since proposition 2.8 shows that $\pi_A \circ \alpha_A$ is an isomorphism, it suffices to prove (2.50) for all W of the form $\pi_A \iota_Z F$, where $\iota_Z F = \alpha_A(a_0, \mathbf{a})$. But then

$$\Psi_* W = (a_0, \mathbf{a}) + O((|a_0| + |\mathbf{a}|) \lambda^{1-\delta})$$

by proposition 2.15, and therefore

$$\begin{aligned} |(\Psi^* h_{\text{cyl}}(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2))| &\leq C\lambda^{1-\delta}(|a_0|^2 + |\mathbf{a}|^2) \\ &\leq C\lambda^{1-\delta}(\Psi^* h_{\text{cyl}})(W, W). \end{aligned} \quad (2.51)$$

On the other hand, we will show that for λ_0 sufficiently small,

$$|g(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2)| \leq \frac{1}{2}\varepsilon(\Psi^* h_{\text{cyl}})(W, W).$$

Actually, the estimate follows by expressing $g(W, W)$ in terms of $\|\iota_Z F\|^2$. By definition, $g(W, W) = \|W\|^2 = \|\pi_A \iota_Z F\|^2$. Since π_A is an orthogonal projection, the Pythagorean theorem for the decomposition $\iota_Z F = \pi_A \iota_Z F + (1 - \pi_A) \iota_Z F$ gives

$$g(W, W) = \|\iota_Z F\|^2 - \|(1 - \pi_A) \iota_Z F\|^2.$$

We now estimate the terms on the right-hand side using proposition 2.5 and proposition 2.8,

- proposition 2.5 provides the asymptotic expansion of the main term,

$$\|\iota_Z F\|^2 = 4\pi^2(2a_0^2 + |\mathbf{a}|^2) + O(\lambda^2(|a_0|^2 + |\mathbf{a}|^2)).$$

- proposition 2.8 bounds the norm of the rejected component, which is of a higher order,

$$\|(1 - \pi_A) \iota_Z F\|^2 = O(\lambda^2(|a_0|^2 + |\mathbf{a}|^2)).$$

Substituting these into the expression for $g(W, W)$ allows us to find the difference from the ideal quadratic form

$$g(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2) = O(\lambda^2(|a_0|^2 + |\mathbf{a}|^2)).$$

Since $\lambda^{1-\delta}$ is the dominant error term as $\lambda \rightarrow 0$, we have

$$|g(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2)| \leq C\lambda^{1-\delta}(|a_0|^2 + |\mathbf{a}|^2).$$

From the previous step in the theorem's proof, we know that $(\Psi^* h_{\text{cyl}})(W, W)$ is asymptotically proportional to $|a_0|^2 + |\mathbf{a}|^2$. We can therefore replace this term in the error bound by $(\Psi^* h_{\text{cyl}})(W, W)$ at the cost of changing the constant

$$|g(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2)| \leq C'\lambda^{1-\delta}(\Psi^* h_{\text{cyl}})(W, W). \quad (2.52)$$

Since $\lambda \rightarrow 0$, for any given $\varepsilon > 0$, we can choose λ_0 sufficiently small such that for all $\lambda < \lambda_0$, the coefficient $C'\lambda^{1-\delta} < \varepsilon/2$. This yields the desired inequality.

Combining (2.51) with (2.52) yields

$$\begin{aligned} |g(W, W) - (\Psi^* h_{\text{cyl}})(W, W)| &\leq |g(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2)| + |(\Psi^* h_{\text{cyl}}(W, W) - 4\pi^2(2a_0^2 + |\mathbf{a}|^2))| \\ &\leq C\lambda^{1-\delta}(\Psi^* h_{\text{cyl}})(W, W). \end{aligned}$$

Therefore taking λ_0 small enough, we obtain

$$|g(W, W) - (\Psi^* h_{\text{cyl}})(W, W)| \leq \varepsilon(\Psi^* h_{\text{cyl}})(W, W),$$

and the result follows. □

Chapter 3 Geometry of Yang-Mills Moduli Space for Definite Manifolds, II: Geometry of the Cones

In this section we will describe the L^2 metric on the moduli space in a neighborhood of a reducible connection. Our approach is to use the fibration $\widetilde{\mathcal{M}} \rightarrow \mathcal{M}$ of Theorem 1.2.7 to reduce the problem to a calculation in finite-dimensional Riemannian geometry.

We will work on a principal $G = \mathrm{SU}(2)$ bundle over a 4-manifold M satisfying $b_1(M) = b_2^+(M) = 0$, but will allow any instanton number $k \geq 1$. Fix a Sobolev norm $s > 1$. By a theorem of Uhlenbeck [8, §3] we have, after perturbing the metric on M if necessary, that $\mathcal{M}^* = \mathcal{M}^*$ (i.e. that $\ker d_A^- = \{0\}$ for all SD A). The moduli spaces $\widetilde{\mathcal{M}}$ and $\mathcal{M}^*$ are then smooth, and $\mathcal{M} \setminus \mathcal{M}^*$ consists of the gauge-equivalence classes of reducible self-dual connections whose holonomy reduces the bundle P to an S^1 -bundle. More specifically, there exists subbundle $Q \subset P$, such that Q is S^1 -bundle (recall the definition of reducible connection: $Z(G) \subsetneq \mathrm{Stab}_g(A) = \{g \in \mathcal{G} \mid g \cdot A = A\}$). The set of such reducible connections is in 1-1 correspondence with $\{u \in H^2(M; \mathbb{Z}) \mid u \cup u = 1\}$, and hence consists of a finite number of points in $\widetilde{\mathcal{M}}$ (cf. Lawson's book, "The theory of gauge fields in four dimensions", section 4.3). Our goal is to describe the geometry of $\widetilde{\mathcal{M}}$ near these points.

The reducible connections can also be characterized in terms of the isometric action of $\mathcal{G}/(\mathcal{G}_0 \times \mathcal{Z}) \cong \mathrm{SU}(2)/\{I, -I\} \cong \mathrm{SO}(3)$ on $\widetilde{\mathcal{M}}$ described in the previous section. The stabilizer at an irreducible point is trivial, so $\widetilde{\mathcal{M}}^* \rightarrow \widetilde{\mathcal{M}}$ is a principal $\mathrm{SO}(3)$ -bundle and $\dim \widetilde{\mathcal{M}} = \dim \mathcal{M} + 3 = 8k$. Before we describe the stabilizer at a reducible point $[A_0] \in \widetilde{\mathcal{M}}$, we need

The adjoint bundle $\mathrm{Ad}P$ associated with an $\mathrm{SU}(2)$ -bundle has fibers isomorphic to $\mathfrak{su}(2)$, which is a 3-dimensional Lie algebra. Since $\mathfrak{su}(2)$ is isomorphic to $\mathbb{R}^3$ with the cross-product structure, we can think of the adjoint bundle as a vector bundle with 3-dimensional fibers. A reducible connection A_0 satisfies $\nabla^{A_0}\Phi = 0$ for some section Φ , which means that Φ is constant under parallel transport, so at each point in M , the fiber of $\mathrm{Ad}P$ contains a distinguished vector Φ , just like picking a fixed axis in $\mathbb{R}^3$.

In order to study the conjugation action of gauge transformations on the fibers of the adjoint bundle $\text{Ad}P$, we introduce a geometric observation that provides a more intuitive perspective. **The adjoint representation of $\text{SU}(2)$ is equivalent to the standard action of $\text{SO}(3)$ on $\mathbb{R}^3$, the fibers of $\text{Ad}P$ behave like copies of $\mathbb{R}^3$ with an $\text{SO}(3)$ -action.** This observation bases on the following fact: there exists a double cover homomorphism $\phi: \text{SU}(2) \rightarrow \text{SO}(3)$, whose kernel is $\{\pm I\}$, and is defined as

$$\phi(U) \cdot v = U \left(\sum_{i=1}^3 v_i \sigma_i \right) U^\dagger,$$

where σ_i are the Pauli matrices, v is a three-dimensional real vector, and $\cdot$ denotes the dot product. The geometric meanings of this mapping contain

- Each element of $\text{SU}(2)$ corresponds to a rotation in $\text{SO}(3)$.
- $\text{SU}(2)$ is the universal covering group of $\text{SO}(3)$.

The Lie algebra $\mathfrak{su}(2)$ can be viewed as the three-dimensional Euclidean space $\mathbb{R}^3$. Specifically, take the basis vectors $T_i = \frac{i}{2} \sigma_i$, where $k = 1, 2, 3$ and σ_i are the Pauli matrices, in this case, $\mathfrak{su}(2)$ is isomorphic to $\mathbb{R}^3$. The geometric meaning of the adjoint action $\text{Ad}_g(X) = gXg^\dagger$, where $g \in \text{SU}(2)$ and $X \in \mathfrak{su}(2)$, can be encapsulated in: under the isomorphism $\mathfrak{su}(2) \cong \mathbb{R}^3$, this adjoint action corresponds to $\phi(g) \in \text{SO}(3)$ rotating the vector $\mathbf{x}$

$$\text{Ad}_g(X) \leftrightarrow \phi(g) \cdot \mathbf{x}.$$

It is worth noting that the presence of the center subgroup of $\text{SU}(2)$, $\{I, -I\}$ ensures that the gauge transformations reduce correctly to $\text{SO}(3)$, removing redundant transformations. In standard $\mathbb{R}^3$ space, the set of transformations that fix a specific vector consists only of rotations around that vector. These rotations form a circle group S^1 , embedded inside $\text{SO}(3)$. Namely, the stabilizer of a vector Φ in $\mathbb{R}^3$ under $\text{SO}(3)$ is:

$$\text{Stab}(\Phi) = \{R \in \text{SO}(3) \mid R\Phi = \Phi\}.$$

These are exactly the rotations around the axis defined by Φ . Thus, the presence of a covariantly constant Φ means that the gauge transformations that preserve A_0 must belong to this S^1 subgroup of $\text{SO}(3)$. In summary, the existence of a parallel section Φ that defines a preferred direction in $\text{Ad}P$, thereby breaking the full rotational symmetry of $\text{SO}(3)$ down to an S^1 subgroup.

For purpose of characterize the set of rotations that preserve the direction of Φ , we need the following lemma:

Lemma 3.1. *If two vectors $u, v \in \mathfrak{su}(2)$ commute, then they are collinear.*

CHAPTER 3 GEOMETRY OF YANG-MILLS MODULI SPACE FOR DEFINITE MANIFOLDS, II: GEOMETRY OF THE CONES

Proof. The Lie algebra $\mathfrak{su}(2)$ consists of all 2×2 anti-Hermitian matrices with trace zero. Its standard basis is given by:

$$T_1 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2} \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}.$$

These basis elements satisfy the commutation relations:

$$[T_i, T_j] = \sum_k \epsilon_{ijk} T_k,$$

where ϵ_{ijk} is the Levi-Civita symbol. Then this two vectors $u, v \in \mathfrak{su}(2)$ can be written as:

$$\begin{aligned} u &= x_1 T_1 + x_2 T_2 + x_3 T_3, & x_1, x_2, x_3 &\in \mathbb{R}. \\ v &= y_1 T_1 + y_2 T_2 + y_3 T_3, & y_1, y_2, y_3 &\in \mathbb{R}. \end{aligned}$$

Using the bilinearity of the Lie bracket, the commutator $[u, v]$ expands as:

$$[u, v] = \sum_{i,j=1}^3 x_i y_j [T_i, T_j] = \sum_{k=1}^3 \left(\sum_{i,j=1}^3 \epsilon_{ijk} x_i y_j \right) T_k.$$

Define $z_k = \sum_{i,j=1}^3 \epsilon_{ijk} x_i y_j$, thus

$$[u, v] = z_1 T_1 + z_2 T_2 + z_3 T_3.$$

If $[u, v] = 0$, then $z_1 = z_2 = z_3 = 0$. This implies:

$$\sum_{i,j=1}^3 \epsilon_{ijk} x_i y_j = 0, \quad \forall k = 1, 2, 3.$$

Let (x_1, x_2, x_3) and (y_1, y_2, y_3) represent the components of vectors $\vec{x}$ and $\vec{y}$, respectively, then above condition implies $\vec{x} \times \vec{y} = 0$. Hence the vectors are collinear, i.e. there exists a scalar $t \in \mathbb{R}$ such that $\vec{x} = t\vec{y}$. Returning to u and v , this implies $u = tv$. □

Therefore, the elements that commute with Φ form the centralizer of Φ in $\mathfrak{su}(2)$, which consists of scalar multiples of Φ . Exponentiating these elements in $SU(2)$ generates the subgroup of rotations around Φ , then under conjugation by e^{tX} , we get $e^{tX} \Phi e^{-tX} = \Phi$, which means e^{tX} does not change its direction—it corresponds to a rotation around the axis defined by Φ . Thus, commutation with Φ precisely characterizes the set of rotations that leave the direction of Φ unchanged.

Since $d|\Phi|^2 = 2(\Phi, \nabla^{A_0} \Phi) = 0$, the length of Φ is preserved under parallel transport, which implies that $|\Phi|^2$ is constant. We normalize this to $|\Phi| = 1$.

A typical element $\Phi \in \mathfrak{su}(2)$ can be written as:

$$\Phi = i\mathbf{n} \cdot \boldsymbol{\sigma} = i(n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3),$$

where σ_i are the Pauli matrices and $\mathbf{n}$ is a unit vector in $\mathbb{R}^3$ specifying a preferred direction. The exponential map gives:

$$\exp(t\Phi) = \exp(it\mathbf{n} \cdot \boldsymbol{\sigma}) = \cos(\mathbf{n} \cdot \boldsymbol{\sigma}t) + i\sin(\mathbf{n} \cdot \boldsymbol{\sigma}t).$$

Using the well-known formula for exponentiating Pauli matrices:

$$e^{i\theta\mathbf{n} \cdot \boldsymbol{\sigma}} = \cos\theta + i(\mathbf{n} \cdot \boldsymbol{\sigma})\sin\theta,$$

we get:

$$\exp(t\Phi) = \cosh(t) + i(\mathbf{n} \cdot \boldsymbol{\sigma})\sinh(t).$$

Since $SU(2)$ represents **compact** rotations, we instead use the formula with i absorbed into Φ :

$$\exp(t\Phi) = \cos(t)I + \sin(t)\Phi.$$

Periodicity of $SU(2)$

Unlike $SO(3)$, which has a fundamental periodicity of 2π , the group $SU(2)$ has a **double cover property**, meaning that:

$$\exp(2\pi\Phi) = -I.$$

This arises because $SU(2)$ has a **periodicity of** 4π instead of 2π , meaning that after rotating by 4π , we return to the identity:

$$\exp(4\pi\Phi) = I.$$

Thus, the elements satisfying

$$\exp(t\Phi) = \pm I$$

occur when t is a multiple of 2π , meaning that $t\Phi$ generates a **circle subgroup** S^1 inside $SU(2)$, corresponding to rotations around Φ in $SO(3)$.

Conclusion: The Stabilizer as an S^1 Subgroup

The stabilizer elements can be written in terms of the exponential map as $\exp(t\Phi)$, $t \in \mathbb{R}$. The periodicity property of $SU(2)$ ensures that $\exp(2\pi\Phi) = -I$, and the elements of the stabilizer satisfy

$$\exp(t\Phi) = \pm I \quad \Leftrightarrow \quad t \text{ is a multiple of } 2\pi.$$

Thus, these transformations form a circle subgroup $S^1 \subset \mathrm{SO}(3)$.

which can be described as follows (notice that S^1 is the only Lie subgroup of $\mathrm{SO}(3)$). This Φ has constant length ($d|\Phi|^2 = 2(\Phi, \nabla\Phi) = 0$) which we normalize to be 1. For convenience we use, in this section only, the metric on $\mathrm{Ad} P$ induced by $-\frac{1}{2}$ the Killing form of $\mathfrak{su}(2)$, so the identity $[u, [v, w]] = (u, w)v - (u, v)w$ holds. For each unit vector $u \in \mathfrak{su}(2)$ we then have $\exp(2\pi u) = -1$, and $\exp(tu) = \pm 1 \Leftrightarrow t$ is a multiple of 2π . (Although our metric here is twice that used in §§3–5 (where, as is customary, our metric is minus the trace form of the standard representation on $\mathbb{C}^2$), all the results of the present section—in particular Theorem 2.8 and Corollary 2.9—are completely independent of the choice of normalization.) For any $t \in \mathbb{R}$, $\exp(t\Phi)$ is a gauge transformation fixing A_0 and $\exp(t\Phi) \in \mathcal{G} \Leftrightarrow t$ is a multiple of 2π . Hence $\exp(t\Phi)$, $0 \leq t \leq 2\pi$, projects to a circle in $\mathcal{G}/(\mathcal{G}_0 \times \mathcal{Z}) = \mathrm{SO}(3)$ which acts isometrically on $\widetilde{\mathcal{M}}$ fixing the point $p = [A_0]$. We now make four observations.

(i) Since $g_t = \exp(t\Phi) \in \mathcal{G}$ takes ∇ to $g_t \circ \nabla \circ g_t^{-1} = \nabla - t\nabla\Phi + O(t^2)$, the circle action generated by Φ on $\mathcal{A}$ is given infinitesimally by the Killing vector field

$$\hat{\xi}(A) = -\nabla^A \Phi. \quad (3.1)$$

The corresponding circle action on the quotient $\widetilde{\mathcal{M}}$ has an infinitesimal generator the Killing field

$$\xi([A]) = \pi_* \hat{\xi}(A).$$

(ii) The differential of this S^1 action at $p = [A_0]$ is the isotropy representation of S^1 on $T_p \widetilde{\mathcal{M}}$. To compute it, first note that the differential of the S^1 action on $\mathcal{A}^s$ at A_0 is given by

$$B \mapsto \frac{d}{ds} g_t(\nabla^{A_0} + sB)g_t^{-1}|_{s=0} = (\mathrm{Ad} g_t)B \quad \text{for } B \in T_{A_0} \mathcal{A}^s.$$

This differential preserves the vertical subspace $\{d_{A_0} X | X \in L_{s+1}^2(\mathrm{Ad} P)\}$ satisfies $X(x_0) = 0$ since $(\mathrm{Ad} g_t)d_A X = d_A((\mathrm{Ad} g_t)X)$ with $((\mathrm{Ad} g_t)X)(x_0) = 0$. It must therefore preserve the L_s^2 orthogonal complement (since S^1 acts isometrically). Thus if we identify $T_p \widetilde{\mathcal{M}}$ with $\widetilde{\mathcal{H}}_A$ as in (1.13), then the isotropy representation is

$$\mathrm{Ad} g_t: \widetilde{\mathcal{H}}_A \rightarrow \widetilde{\mathcal{H}}_A. \quad (3.2)$$

(iii) Writing $\mathrm{Ad} g_t = \exp(tJ)$, where $J = \mathrm{ad} \Phi$, we see that the infinitesimal generator of the isotropy representation is

$$J = [\Phi, \cdot]: \widetilde{\mathcal{H}}_A \rightarrow \widetilde{\mathcal{H}}_A. \quad (3.3)$$

(iv) The infinitesimal isotropy representation can also be described in terms of the local Riemannian geometry of $\widetilde{\mathcal{M}}$ by linearizing the Killing vector field ξ at p as follows.

CHAPTER 3 GEOMETRY OF YANG-MILLS MODULI SPACE FOR DEFINITE MANIFOLDS, II: GEOMETRY OF THE CONES

Let $\{\Psi_s\}$ be the flow of ξ , choose $B \in T_p \widetilde{\mathcal{M}}$, and extend B arbitrarily to a neighborhood of p . Then since $\xi(p) = 0$,

$$\frac{d}{dt}[(\Psi_s)_* B]_p = (\Psi_s)_*(-L_\xi B)|_p = (\Psi_s)_*(\nabla_B \xi - \nabla_\xi B)|_p \quad (3.4)$$

$$= (\Psi_s)_* p(JB), \quad (3.5)$$

when J is defined by $J(B) = (\nabla_B^\xi \xi)(p)$ (here ∇^ξ is the Levi-Civita connection on $\widetilde{\mathcal{M}}$). It then follows from the uniqueness theorem for ordinary differential equations that $(\Psi_s)_* p = \exp(sJ)$. Thus (2.3) is alternatively given by

$$J = \nabla^\xi \xi \in \text{End}(T_p \widetilde{\mathcal{M}}). \quad (3.6)$$

(Indeed, one can check directly that (2.3) follows from (2.1) and (2.5).) Note that since the isotropy representation is orthogonal, J is skew-symmetric.

These observations reduce the original gauge theory problem to the following problem in pure Riemannian geometry. We are given an $8k$ -dimensional Riemannian manifold $\widetilde{W}$ ($\widetilde{\mathcal{M}}$ in our application) on which $\text{SO}(3)$ acts isometrically. We assume that the action is free except along a finite number of exceptional orbits $\{\mathcal{O}_i \subset \widetilde{W}\}$ at each point of which the isotropy subgroup is S^1 (so each $\mathcal{O}_i$ is diffeomorphic to $\text{SO}(3)/S_1 = S_2$). Let $\pi: \widetilde{W} \rightarrow W$ denote the projection onto the orbit space and let $x_i = \pi(\mathcal{O}_i)$. Then there is a Riemannian metric on $W^* = W - \{x_i\}$ such that $\pi: \widetilde{W}^* = \widetilde{W} - \{\mathcal{O}_i\} \rightarrow W^*$ is a Riemannian submersion. We seek a description of the geometry of W near each x_i .

This geometry problem is solved by Theorem 2.8 below. Our answer will make use of the following geometric quantities. The infinitesimal $\text{SO}(3)$ -action is a linear map L which associates to each $v \in \text{SO}(3)$ a Killing vector field $L(v)$ on W ; we will often write ξ_v for $L(v)$. The pointwise adjoint of L is an $\mathfrak{so}(3)$ -valued 1-form L^* on W . For each $x \in W$, the operator $L^*L \in \text{End}(\mathfrak{so}(3))$ is selfadjoint and nonnegative; it is strictly positive off $\bigcup \mathcal{O}_i$ and has a (simple) zero eigenvalue at each point $x \in \mathcal{O}_i$. (In our gauge theory problem L is the restriction of d_A to the Lie algebra $\mathfrak{g} = \mathfrak{so}(3) = \{(\square_A^0)^{-s} G_A^s\}$, and L^*L is the restriction to $\mathfrak{g}$ of the Laplacian $d_A^* d_A$ of the fundamental elliptic complex (0.1).) The fact that the smallest eigenvalue of L^*L approaches zero as x approaches $\mathcal{O}_i$ will be central to the discussion below.

Fix a point $p \in \widetilde{W}$ on an exceptional orbit $\mathcal{O} = \pi^{-1}(x_0)$. Let $v_p \in \mathfrak{so}(3)$ be a unit-length generator of its isotropy subgroup $S_p^1 \subset \text{SO}(3)$. Then the infinitesimal action of S_p^1 is the Killing vector field $\xi = L(v_p)$ and, as above, the isotropy representation is generated by $J = \nabla \xi$. The local topology around $\mathcal{O}$ is described by the following well-known theorem.

Theorem 3.2 (Differentiable Slice Theorem). *Let G be a compact Lie group acting isometrically on a finite-dimensional Riemannian manifold $\widetilde{W}$. Let G_p be the isotropy subgroup of a point $p \in \widetilde{W}$ with orbit $\mathcal{O}_p$ (so G_p acts on the normal bundle $N_p \mathcal{O}$ by the isotropy*

representation). Then the exponential map is an equivariant diffeomorphism from $N_p\mathcal{O}$ to a neighborhood of $\mathcal{O}_p$ in $\widetilde{\mathcal{W}}$.

This theorem is a simple consequence of the uniqueness theorem for solutions to ordinary differential equations.

To apply this we must first identify the isotropy representation of our $\mathrm{SO}(3)$ action.

Lemma 3.3. $J^2 = -\mathrm{Id}$. Hence $\dim \widetilde{\mathcal{W}}$ is even and $T_p\widetilde{\mathcal{W}} = T_p\mathcal{O} \oplus N_p\mathcal{O} \cong \mathbb{C} \oplus \mathbb{C}^{4k-1}$ as S_p^1 -representations.

Remark. In our gauge theory application, Lemma 2.1 can also be proved by entirely analytic methods. This is done in the appendix.

We now describe the local structure of the fibration $\widetilde{\mathcal{W}} \rightarrow \mathcal{W}$ near $\mathcal{O}$. For this, we use polar coordinates to identify the normal space $N_p\mathcal{O} - \{p\}$ with $(0, \infty) \times \Sigma_p$, where $\Sigma_p \cong S^{8k-3}$ is the unit sphere. The Differentiable Slice Theorem then implies that the exponential map

$$\widetilde{F}_p = \exp_p: (0, \varepsilon) \times \Sigma_p \rightarrow \widetilde{U}^* \quad (3.7)$$

$$= \{\exp_p X | X \in N_p\mathcal{O}, 0 < |X| < \varepsilon\} \subset \widetilde{\mathcal{W}}^* \quad (3.8)$$

Theorem 3.4. Let (M, g) be a compact oriented 1-connected 4-manifold with positive-definite intersection form, and let $(\mathcal{M}_k, g)$ be the moduli space of self-dual $\mathrm{SU}(2)$ connections on the bundle P over M with instanton number $k = \frac{1}{4}p_1(\mathrm{Ad}P) \geq 1$ (with its L^2 metric). Fix a reducible connection $[A] \in \mathcal{M}_k$. Let g_0 be the standard metric on $\mathbb{P} = \mathbb{CP}^{4k-2}$. Then there are a number r_0 , a neighborhood U of $[A]$ in $\mathcal{M}_k$, and a diffeomorphism

$$F: (0, r_0) \times \mathbb{P} \rightarrow U - \{[A]\}, \quad (3.9)$$

which extends to a homeomorphism from the open cone $[0, r_0) \times \mathbb{P}$ to U . In this coordinate system: (a) The metric satisfies

$$F^*g = dr^2 \oplus r^2 (g_0 + O(r^2)) \quad (3.10)$$

(b) As $r \rightarrow 0$, the sectional curvatures σ of F^*g satisfy (0.3)

$$\begin{aligned} \sigma\left(\frac{\partial}{\partial r}, X\right) &= O(1), \\ \sigma(X, Y) &= \frac{3}{r^2}g_0(JX, Y) + O(1), \end{aligned} \quad (3.11)$$

where $X, Y \in T\mathbb{P}$ and J is the complex structure on $\mathbb{P}$.

Chapter 4 Curvature of Yang-Mills Moduli Spaces

1 Curvature Near the Boundary

Definition 4.1. Let R and W^- be the scalar curvature and anti-self-dual Weyl tensor of (M, g) . Define an operator $\mathcal{W}^-$ on two-forms by $\mathcal{W}^-(\theta^i \wedge \theta^j) = \frac{1}{2}(W^-)^{ij}_{kl} \theta^k \wedge \theta^l$; this preserves the subspace of anti-self-dual two forms. We say that (M, g) satisfies the Λ_-^2 curvature condition at $p \in M$ if the operator

$$\frac{R}{6} \text{Id} - 2\mathcal{W}^- : \bigwedge_{-}^2 T_p^* M \rightarrow \bigwedge_{-}^2 T_p^* M$$

is positive-definite.

Theorem 4.2. Let (M, g) be a compact, simply connected, oriented, positive-definite 4-manifold, and $\mathcal{M}$ the space of 1-instantons over M . Assume that (M, g) satisfies the Λ_-^2 curvature condition. Then for λ_0 sufficiently small the sectional curvature of $(\mathcal{M}_{\lambda_0}, g)$ is bounded above and below.

Theorem 4.3. Assume all hypotheses of Theorem 1.1 except the Λ_-^2 curvature condition. Then $\text{Riem}_M^{\text{tan}}$ extends continuously to $\mathcal{M}_{\lambda_0}$, and the limit depends only on Riem_M . Specifically,

$$\text{Riem}_M^{\text{tan}}|_{\partial \mathcal{M}} := \lim_{\lambda \rightarrow 0} j_{\lambda}^* \text{Riem}_M^{\text{tan}} = L(\text{Riem}_M)$$

for some geometrically natural endomorphism L .

Our approach to proving Theorems 1.4.2 and theorem 1.4.3 will rely on the expression derived in [GP1] for the sectional curvature of $\mathcal{M}$ at a point $[A]$ represented by a self-dual connection A . To review this formula, recall that the tangent space $T_{[A]}\mathcal{M}$ is naturally isomorphic to the harmonic space

$$H_A = \{X \in \Omega^1(\text{Ad}P) \mid d^{A*}X = 0 \in \Omega^0(\text{Ad}P) \text{ and } d_-^A X = 0 \in \Omega_-^2(\text{Ad}P)\}, \quad (4.1)$$

where d^{A*} is the adjoint of the covariant exterior derivative d^A and d_-^A is the anti-self-dual part of d^A .

2 Sharp Decay Estimates for Yang-Mills Fields

Theorem 4.4. *Let (M, g) be a Riemannian manifold of bounded geometry and let $P \rightarrow M$ be a principal G -bundle with G compact. There exist constants $C, \varepsilon_0, \rho > 0$, depending on (M, g) and G , such that if $0 \leq 4r_0 \leq r_1 \leq \rho$ and A is a Yang-Mills connection on $\Omega = \Omega(p, r_0, r_1)$ with $\|F_A\|_\Omega^2 \leq \varepsilon_0$, then*

$$|F_A(y)| \leq C \left(\frac{r_0^2}{r^4} + \frac{1}{r_1^2} \right) \|F_A\|_\Omega \quad \text{for } y \in \Omega(p, 2r_0, r_1/2). \quad (4.2)$$

Moreover, if A is a SD or ASD unitary connection on $B(p, r_1)$ with $\|F_A\|_\Omega^2 \leq \varepsilon_0$ then

- If $k \leq E(B(p, r_0)) < k + \frac{1}{2}$ for some $k \in \mathbb{Z}$ then

$$|F_A(y)| \leq \frac{C}{r_1^2} \|F_A\|_\Omega \quad \text{for } y \in \Omega(p, 2r_0, r_1/2). \quad (4.3)$$

- If $k - \frac{1}{2} < E(B(p, r_1)) \leq k$ for some $k \in \mathbb{Z}$ then

$$|F_A(y)| \leq C \frac{r_0^2}{r^4} \|F_A\|_\Omega \quad \text{for } y \in \Omega(p, 2r_0, r_1/2). \quad (4.4)$$

Theorem 4.5. *Let (M, g) be an asymptotically flat 4-manifold, and let G, P be as in Theorem 1.1. There exists constants $C, \epsilon_0 > 0$ and $R \geq R_0$, depending on $C(g)$ and G , such that if $4R_0 \leq 4r_0 \leq r_1 \leq \infty$ and A is a Yang-Mills connection on $\Omega = \Omega(r_0, r_1)$ with $\|F\|_\Omega^2 \leq \epsilon_0$ then*

$$|F(y)| \leq C \left(\frac{r_0^2}{r^4} + \frac{1}{r_1^2} \right) \|F\|_\Omega \quad \text{for } y \in \Omega(2r_0, r_1/2). \quad (4.5)$$

Moreover, if A is a SD or ASD unitary connection on $\Omega(r_0, \infty)$ with $\|F_A\|_\Omega^2 \leq \epsilon_0$ then

- If $k \leq E(r_1, \infty) < k + \frac{1}{2}$ for some $k \in \mathbb{Z}$ then

$$|F(y)| \leq C \frac{r_0^2}{r^4} \|F\|_\Omega \quad \text{for } y \in \Omega(2r_0, r_1/2). \quad (4.6)$$

- If $k - \frac{1}{2} < E(r_0, \infty) \leq k$ for some $k \in \mathbb{Z}$ then

$$|F(y)| \leq \frac{C}{r_1^2} \|F\|_\Omega \quad \text{for } y \in \Omega(2r_0, r_1/2). \quad (4.7)$$